

MTH321
Numerical Analysis Of Partial Differential Equations

Numerical Analysis of the Robertson Stiff Chemical Kinetics System

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September 16, 2026

Contents

Contents	1
1 Introduction	2
1.1 Background and Motivation	2
1.2 Problem Statement and Report Roadmap	2
2 Mathematical Formulation	2
2.1 Reaction Kinetics Model	2
2.2 Conservation Invariant and Structural Zero Eigenvalue	3
2.3 Reduced Jacobian and Stiffness Ratio Definition	4
3 Numerical Integration Methods	6
3.1 Explicit Euler Method	6
3.2 Explicit Fourth-Order Runge–Kutta Method (RK4)	7
3.3 Implicit Euler Method with Newton–Raphson Iteration	9
3.4 Trapezoidal Method (Crank–Nicolson) as a Second Implicit Scheme	11
3.5 Adaptive Step-Size Control	12
3.5.1 Common step-doubling controller	13
3.5.2 Adaptive Explicit Euler	13
3.5.3 Adaptive Implicit Euler and local extrapolation	15
4 Linear Stability Analysis	16
4.1 Absolute Stability Regions of the Four Methods	16
4.2 Eigenvalue Analysis of the Robertson System	18
4.3 Step-Size Constraints: Theory versus Experiment	20
5 Numerical Experiments and Validation	22
5.1 Reference Solution Construction	22
5.2 Sanity Checks	23
5.3 Convergence Order Verification	23
5.4 Explicit Euler: Stability Boundary and Physical Admissibility	25
5.5 Solver Diagnostics	26
5.5.1 Newton-Tolerance Sensitivity and Initial Guess	26
5.5.2 Adaptive Step-Size Control	27
5.6 Efficiency Comparison	29
6 Discussion	32
6.1 Physical Interpretation via Quasi-Steady-State Approximation (QSSA)	32
6.2 Why Explicit Methods Fail and Implicit Methods Succeed	36
6.3 Three-Fold Validation Strategy Recap	37
6.4 Limitations	37
7 Conclusions	38
8 References	39
9 Appendix	41

1 Introduction

1.1 Background and Motivation

Stiff initial value problems arise wherever processes with very different rates are coupled, most prominently in chemical kinetics. Their difficulty is not accuracy but stability: since Curtiss and Hirschfelder [1], it has been known that an explicit method may be forced to use a step size determined by the fastest transient, even after that transient has largely decayed. The reaction system introduced by Robertson [2] has become a standard benchmark for stiff ODE solvers. Its three species react through steps whose rate constants differ by a factor $k_2/k_1 = 7.5 \times 10^8$, so that the intermediate species is produced slowly but consumed almost instantly.

Beyond being a benchmark, the problem is a good test case for *validation*, for three reasons. First, no closed-form solution is available on the interval of interest, so accuracy must be established against an independently verified numerical reference. Second, the exact solution obeys a linear invariant and must stay non-negative, which gives diagnostics that are independent of accuracy and can fail independently of it. Third, the step size at which explicit methods break down can be predicted from the Jacobian, which allows a genuine prediction–observation comparison. We therefore treat the problem as a consultant would: the aim is not to reproduce a textbook example, but to produce a validated numerical answer and to show, quantitatively, why it can be trusted.

1.2 Problem Statement and Report Roadmap

We study the initial value problem for the concentrations of the three species, the coefficients of Eq. (4) follow the mass-action convention fixed by the problem statement, without an extra factor of two in the quadratic term. Since $\mathbf{1}^T \mathbf{f} \equiv 0$, the exact solution stays on the plane $y_1 + y_2 + y_3 = 1$ and is non-negative.

The report answers three questions. (i) Are the numerical solutions correct, i.e. do the implementations reproduce their formal orders against a verified reference? (ii) Which step-size restriction is active for each method, accuracy or stability, and does the solution remain physically admissible? (iii) What does each method cost at comparable accuracy? To answer them we implement Explicit Euler, RK4, Implicit Euler and Trapezoidal/Crank–Nicolson (with Newton’s method) and step-doubling adaptive Explicit and Implicit Euler methods (including a locally extrapolated implicit variant), derive their stability properties, and validate them against a tight Radau reference. In short, we find that stiffness appears here as a *stability* restriction, $h < 5.895 \times 10^{-4}$ for Explicit Euler, and that the implicit advantage depends on the accuracy demanded.

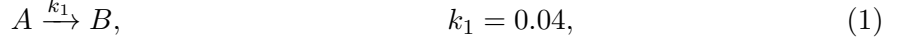
The report is organised as follows. Section 2 presents the model, its invariant and its Jacobian. Section 3 describes the numerical methods, including the Newton iteration and the adaptive controller. Section 4 derives the stability regions, the eigenvalue structure and the step-size constraints. Section 5 reports the experiments: reference construction, convergence orders, stability and admissibility, solver diagnostics and efficiency. Section 6 interprets the results, through a quasi-steady-state analysis (Section 6.1), the explicit-versus-implicit comparison (Section 6.2), a recap of the validation strategy (Section 6.3) and the limitations (Section 6.4). Section 7 concludes.

2 Mathematical Formulation

2.1 Reaction Kinetics Model

We consider the classical stiff chemical kinetics problem due to Robertson [2], which models an autocatalytic reaction network among three chemical species A , B , and C . The scheme consists

of three elementary steps operating on widely separated time scales:



with rate constants spanning approximately eight orders of magnitude (k_1 to k_2). This disparity in reaction rates is the source of the stiffness analyzed in Section 4.

The reaction labels above are mnemonic descriptions of the underlying chemistry; the governing ordinary differential equation system is the authoritative model and fixes the mass-action coefficient convention used throughout this report. With $y = (y_1, y_2, y_3)^\top$ denoting the concentrations of A , B , and C respectively,

$$\begin{aligned} y_1' &= -0.04 y_1 + 10^4 y_2 y_3, \\ y_2' &= 0.04 y_1 - 10^4 y_2 y_3 - 3 \times 10^7 y_2^2, \\ y_3' &= 3 \times 10^7 y_2^2, \end{aligned} \quad (4)$$

subject to the initial condition $y(0) = (1, 0, 0)^\top$.

Each term admits a direct kinetic interpretation: the $-0.04 y_1$ / $+0.04 y_1$ pair reflects the depletion of A and production of B by reaction 1; the $10^4 y_2 y_3$ pair reflects the regeneration of A from B at the expense of B via reaction 3 (with C acting as a catalyst, left unchanged); and the $3 \times 10^7 y_2^2$ terms reflect the self-reaction of B via reaction 2, which both consumes B and produces C .

2.2 Conservation Invariant and Structural Zero Eigenvalue

The mathematical foundation of the Robertson chemical kinetics model rests on an exact linear mass conservation law and an intrinsic structural degeneracy of its governing vector field. Prior to performing any numerical integration or dynamic reduction, we establish these fundamental algebraic properties formally.

Proposition 1 (Mass Conservation and Structural Zero Eigenvalue). *Let $f(y) = [f_1(y), f_2(y), f_3(y)]^\top : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the vector field of the Robertson reaction system defined by*

$$f_1(y) = -k_1 y_1 + k_3 y_2 y_3, \quad (5a)$$

$$f_2(y) = k_1 y_1 - k_3 y_2 y_3 - k_2 y_2^2, \quad (5b)$$

$$f_3(y) = k_2 y_2^2, \quad (5c)$$

where $k_1 = 0.04$, $k_2 = 3 \times 10^7$, and $k_3 = 10^4$. Let $\mathbf{1} = [1, 1, 1]^\top \in \mathbb{R}^3$ denote the ones vector. Then:

1. **Algebraic Identity:** The vector field satisfies $\mathbf{1}^\top f(y) = 0$ identically for all state vectors $y \in \mathbb{R}^3$.
2. **Exact Conservation Law:** Any exact trajectory $y(t)$ subject to initial condition $y(0) = [1, 0, 0]^\top$ remains confined to the two-dimensional invariant simplex hyper-plane

$$\mathcal{M} = \{y \in \mathbb{R}^3 \mid y_1 + y_2 + y_3 = 1\} \quad \forall t \geq 0. \quad (6)$$

3. **Structural Zero Eigenvalue:** The full 3×3 Jacobian matrix $J_f(y) \equiv \frac{\partial f}{\partial y}(y)$ possesses a zero eigenvalue $\lambda_3 = 0$ with associated constant left eigenvector $\mathbf{1}^\top$ everywhere in state space, independent of y or time t .

Proof. We prove each statement sequentially:

1. Algebraic Identity: For an arbitrary vector $y = [y_1, y_2, y_3]^T \in \mathbb{R}^3$, expanding the inner product $\mathbf{1}^T f(y) = f_1(y) + f_2(y) + f_3(y)$ yields

$$\mathbf{1}^T f(y) = (-k_1 y_1 + k_3 y_2 y_3) + (k_1 y_1 - k_3 y_2 y_3 - k_2 y_2^2) + (k_2 y_2^2). \quad (7)$$

Pairwise cancellation of identical reaction terms gives:

$$\mathbf{1}^T f(y) = \underbrace{(-k_1 y_1 + k_1 y_1)}_{=0} + \underbrace{(k_3 y_2 y_3 - k_3 y_2 y_3)}_{=0} + \underbrace{(-k_2 y_2^2 + k_2 y_2^2)}_{=0} = 0. \quad (8)$$

Because this cancellation depends solely on the stoichiometric coefficients of the reaction network and holds without assuming y satisfies the differential equations or non-negativity constraints, $\mathbf{1}^T f(y) \equiv 0$ is a global algebraic identity over $\mathbb{R}^3$.

2. Exact Conservation Law: Along any solution trajectory $y(t)$ governed by $\frac{dy}{dt} = f(y(t))$, the time derivative of the total chemical species sum is

$$\frac{d}{dt} (y_1(t) + y_2(t) + y_3(t)) = \frac{d}{dt} (\mathbf{1}^T y(t)) = \mathbf{1}^T \frac{dy}{dt} = \mathbf{1}^T f(y(t)) \equiv 0. \quad (9)$$

Integrating from $t = 0$ to t gives $\mathbf{1}^T y(t) = \mathbf{1}^T y(0)$. Substituting the benchmark initial condition $y(0) = [1, 0, 0]^T$ yields $\mathbf{1}^T y(0) = 1 + 0 + 0 = 1$. Hence, $y_1(t) + y_2(t) + y_3(t) \equiv 1$ holds for all $t \geq 0$.

3. Structural Zero Eigenvalue: Differentiating the algebraic identity $\mathbf{1}^T f(y) = 0$ with respect to the state vector y via the multivariate chain rule gives

$$\frac{\partial}{\partial y} (\mathbf{1}^T f(y)) = \mathbf{1}^T \frac{\partial f}{\partial y}(y) = \mathbf{1}^T J_f(y) = \mathbf{0}^T. \quad (10)$$

Transposing Eq. (10) yields $J_f(y)^T \mathbf{1} = \mathbf{0} = \mathbf{0} \cdot \mathbf{1}$. Because $\mathbf{1} \neq \mathbf{0}$, this confirms that $\lambda = 0$ is an exact eigenvalue of $J_f(y)^T$ (and consequently of $J_f(y)$) for all $y \in \mathbb{R}^3$. Crucially, this zero eigenvalue is *structural*: it originates entirely from mass conservation embedded in the network stoichiometry rather than local state-dependent numerical coincidences. $\square$

Implications for Stiffness Diagnostic Metrics The existence of a persistent structural zero eigenvalue directly impacts standard numerical stiffness diagnostics. The conventional stiffness ratio $S(t)$ for an ODE system is defined by the spread of Jacobian eigenvalues:

$$S_{\text{naive}}(t) = \frac{\max_j |\operatorname{Re}(\lambda_j(t))|}{\min_j |\operatorname{Re}(\lambda_j(t))|}. \quad (11)$$

If evaluated directly on the full 3×3 Jacobian matrix $J_f(y)$, Proposition 1 dictates that $\min_j |\operatorname{Re}(\lambda_j(t))| \equiv \lambda_3 = 0$ at all times. Substituting this into Eq. (11) results in a division-by-zero defect, rendering $S_{\text{naive}}(t)$ ill-defined or infinitely large regardless of the actual dynamic stiffness. This mathematical singularity underscores the necessity of eliminating the redundant degree of freedom prior to diagnostic evaluation, leading directly to the state-space reduction method formulated in Section 2.3.

2.3 Reduced Jacobian and Stiffness Ratio Definition

Proposition before shows that the exact trajectory is confined to the invariant manifold $M = \{y \in \mathbb{R}^3 : y_1 + y_2 + y_3 = 1\}$ and that the full 3×3 Jacobian carries a structural zero eigenvalue. This section makes that reduction explicit: we eliminate y_3 , derive the resulting 2×2 Jacobian J_r , and prove that its eigenvalues coincide exactly with the two non-trivial eigenvalues of J_f . This J_r is then the object evaluated numerically in Section 4.2 and related to the full analytic Jacobian used by the Newton solver in Sections 3.3–3.4.

Reduced two-dimensional system. Substituting $y_3 = 1 - y_1 - y_2$ into f_1 and f_2 from Eqs. 5a-5b defines a self-contained system on $\mathbb{R}^2$,

$$\dot{y}_1 = g_1(y_1, y_2), \quad \dot{y}_2 = g_2(y_1, y_2), \quad (12)$$

with

$$g_1(y_1, y_2) = -k_1 y_1 + k_3 y_2 (1 - y_1 - y_2), \quad (13)$$

$$g_2(y_1, y_2) = k_1 y_1 - k_3 y_2 (1 - y_1 - y_2) - k_2 y_2^2. \quad (14)$$

Since $g_1(y_1, y_2) = f_1(y_1, y_2, 1 - y_1 - y_2)$ and likewise for g_2 , this is not a separate model: it is f restricted to M and expressed in the affine chart $\iota(y_1, y_2) = (y_1, y_2, 1 - y_1 - y_2)$, which is a smooth bijection from $\mathbb{R}^2$ onto M with smooth inverse $\pi(y_1, y_2, y_3) = (y_1, y_2)$.

Proposition 2 (Reduced Jacobian). *Writing $y_3 = 1 - y_1 - y_2$, the Jacobian of the reduced system (12) is*

$$J_r(y_1, y_2) = \begin{pmatrix} -k_1 - k_3 y_2 & k_3(y_3 - y_2) \\ k_1 + k_3 y_2 & -k_3(y_3 - y_2) - 2k_2 y_2 \end{pmatrix}. \quad (15)$$

Its trace and determinant reduce to

$$\text{tr } J_r = -k_1 - k_3 y_3 - 2k_2 y_2, \quad \det J_r = 2k_2 y_2 (k_1 + k_3 y_2), \quad (16)$$

using the identity $(J_r)_{11} = -(J_r)_{21}$, which itself follows from $g_1 + g_2 = -f_3(y_1, y_2, y_3) = -k_2 y_2^2$ being independent of y_1 .

Proof. Direct differentiation of g_1, g_2 with respect to y_1, y_2 , or equivalently the chain rule $Dg = D\pi \cdot J_f(\iota(y_1, y_2)) \cdot D\iota$ with $D\iota = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}$, gives Eq. (15). The trace/determinant simplification follows by direct expansion and use of $(J_r)_{11} + (J_r)_{21} = 0$. $\square$

Exact reduction, not approximate elimination. It remains to show that the two eigenvalues of J_r are exactly the two non-trivial eigenvalues of the full 3×3 Jacobian J_f — i.e. that eliminating y_3 is an exact restriction to the invariant tangent space, not a heuristic dimension reduction.

Proposition 3 (Exact Correspondence of Spectra). *Let $T = \{v \in \mathbb{R}^3 : \mathbf{1}^T v = 0\}$ be the tangent space of M , and let $\Phi : T \rightarrow \mathbb{R}^2$, $\Phi(v_1, v_2, v_3) = (v_1, v_2)$, be the linear isomorphism induced by π . Then for every $y \in \mathcal{M}$ (i.e. $y_3 = 1 - y_1 - y_2$),*

$$\text{spec } J_f(y) = \{0\} \cup \text{spec } J_r(\pi(y)) \quad (17)$$

where J_r is evaluated at $\pi(y) = (y_1, y_2)$ using Eq. (15) with $y_3 = 1 - y_1 - y_2$.

Proof. Differentiating the algebraic identity $\mathbf{1}^T f(y) \equiv 0$ (Proposition 1, part 1) with respect to y gives $\mathbf{1}^T J_f(y) \equiv 0$ for every $y \in \mathbb{R}^3$; that is, $\mathbf{1}$ is a left null vector of $J_f(y)$, equivalently $\text{im } J_f(y) \subseteq T$ for every input, not only for inputs already in T . Choosing a basis of $\mathbb{R}^3$ adapted to the splitting $\mathbb{R}^3 = T \oplus \text{span}\{e\}$ (any $e \notin T$), the image condition forces $J_f(y)$ into the block-triangular form

$$J_f(y) \sim \begin{pmatrix} A & b \\ 0 & 0 \end{pmatrix},$$

where A is the 2×2 matrix representing $J_f(y)|_T : T \rightarrow T$ in that basis. Hence $\text{spec}(J_f(y)) = \{0\} \cup \text{spec}(A)$, recovering the structural zero eigenvalue of Proposition 1, part 3, together with the two remaining eigenvalues carried by A .

It remains to identify A with J_r . Because $\text{im } J_f(y) \subseteq T$ in particular for inputs drawn from T , the restriction $J_f(y)|_T : T \rightarrow T$ is a well-defined linear operator, and since Φ is a linear isomorphism $T \rightarrow \mathbb{R}^2$, A is similar to $\Phi \circ J_f(y)|_T \circ \Phi^{-1}$ regardless of which basis of T was used above. Direct computation shows $\Phi \circ J_f(y)|_T \circ \Phi^{-1} = D\pi \cdot J_f(y) \cdot D\iota = J_r(\pi(y))$, i.e. exactly the matrix of Proposition 2. Similar matrices share eigenvalues, so $\text{spec}(A) = \text{spec}(J_r(\pi(y)))$, which gives Eq. (17). $\square$

For $y \notin \mathcal{M}$ the matrix $D\pi J_f(y) D\iota$ equals Eq. (15) evaluated at the actual y_3 , not at $1 - y_1 - y_2$; since all uses below are on $\mathcal{M}$, the restriction costs nothing. Proposition 3 justifies working with J_r throughout the remainder of the report: its two eigenvalues are the two non-trivial eigenvalues of the full system, exactly, at every point on M — not an approximation obtained by discarding a row and column.

Stiffness ratio.

Definition 1 (Stiffness Ratio). *Let $\lambda_1(t), \lambda_2(t)$ denote the eigenvalues of $J_r(y_1(t), y_2(t))$ along a trajectory. Provided both eigenvalues satisfy $|\lambda_j(t)| > \varepsilon_\lambda$ for a fixed numerical threshold ε_λ separating a genuinely nonzero eigenvalue from floating-point noise, define*

$$S(t) = \frac{\max_j |\text{Re } \lambda_j(t)|}{\min_j |\text{Re } \lambda_j(t)|}. \quad (18)$$

The threshold ε_λ is needed because, from Eq. (16), $\det J_r = 2k_2y_2(k_1 + k_3y_2)$ vanishes exactly whenever $y_2 = 0$ — in particular at the initial condition $y_2(0) = 0$, where one eigenvalue of J_r is exactly zero and $S(t)$ is undefined. This is not a numerical artifact but an exact algebraic degeneracy at $t = 0$; accordingly, $S(t)$ is only plotted from the first positive output time $t = 10^{-8}$ onward, and ε_λ is calibrated in Section 4.2 once $\lambda_1(t), \lambda_2(t)$ have been evaluated numerically along the reference trajectory.

Interface to later sections. The closed-form J_r derived here serves two roles beyond the present section: (i) Section 4.2 evaluates $\lambda_{1,2}(t)$ and $S(t)$ from Eq. (16) along the reference trajectory of Section 5.1; and (ii) Newton’s method for implicit Euler and trapezoidal (Sections 3.3–3.4) is applied to the full three-dimensional stage equation, with iteration matrix $I - hJ(Y)$ built from the full Jacobian J of Section 2.1, not from J_r directly. Section 3.3 shows that J_r is nonetheless the exact restriction of J to the invariant tangent space (Proposition 3), so the two matrices carry the same spectral information: any error in J_r would still be visible in the stiffness diagnostics of Section 4.2, even though the Newton solve itself uses J , not J_r .

3 Numerical Integration Methods

3.1 Explicit Euler Method

The explicit (forward) Euler method is the simplest one-step scheme for approximating the solution of an initial value problem [3, 4].

$$y'(t) = f(t, y(t)), \quad y(t_0) = y_0, \quad (19)$$

where, for the Robertson system, $y = (y_1, y_2, y_3)^\top$ and f is the right-hand side defined in Section 2.1. Given a step size $h > 0$ and a discrete time grid $t_n = t_0 + nh$, the method is obtained by replacing the derivative in (19) with a forward difference quotient, yielding the explicit update formula

$$y_{n+1} = y_n + h f(t_n, y_n), \quad n = 0, 1, 2, \dots \quad (20)$$

Because the right-hand side of (20) depends only on the already-known state y_n , no equation needs to be solved at each step; the new approximation y_{n+1} is computed directly. This non-implicit character is the defining feature of the method and the source of both its simplicity and its stability limitations, discussed in Section 4.

Local truncation error Expanding the exact solution $y(t_{n+1})$ in a Taylor series about t_n ,

$$y(t_{n+1}) = y(t_n) + h y'(t_n) + \frac{h^2}{2} y''(t_n) + \mathcal{O}(h^3), \quad (21)$$

and comparing with the update rule (20), the local truncation error (the error introduced in a single step, assuming the previous value y_n were exact) is

$$\tau_n = y(t_{n+1}) - y(t_n) - h f(t_n, y(t_n)) = \frac{h^2}{2} y''(t_n) + \mathcal{O}(h^3) = \mathcal{O}(h^2). \quad (22)$$

Since the local error is $\mathcal{O}(h^2)$ and accumulates over $\mathcal{O}(1/h)$ steps on a fixed interval $[t_0, T]$, the global error satisfies

$$\max_n \|y(t_n) - y_n\| = \mathcal{O}(h), \quad (23)$$

so explicit Euler is a first-order method. This low order motivates the introduction of the fourth-order Runge–Kutta method in Section 3.2 as a higher-accuracy alternative for the non-stiff regimes of the problem.

Application to the Robertson system For the Robertson system, (20) is applied componentwise to $y = (y_1, y_2, y_3)^\top$ with f as given in (5). Because the three reaction rate constants span roughly eight orders of magnitude, the fastest component of f dictates an extremely restrictive upper bound on h for (20) to remain numerically stable; this bound is derived quantitatively from the eigenvalues of the (reduced) Jacobian in Section 4.3. In the numerical experiments of Section 5, explicit Euler therefore serves primarily as a baseline that illustrates the practical consequences of stiffness, rather than as a competitive integrator for this problem.

3.2 Explicit Fourth-Order Runge–Kutta Method (RK4)

Method definition. The classical four-stage explicit Runge–Kutta method (RK4) is defined by the Butcher tableau

$$\begin{array}{c|cccc} 0 & & & & \\ 1/2 & 1/2 & & & \\ 1/2 & 0 & 1/2 & & \\ 1 & 0 & 0 & 1 & \\ \hline & 1/6 & 1/3 & 1/3 & 1/6 \end{array} \quad (24)$$

i.e. $c = (0, \frac{1}{2}, \frac{1}{2}, 1)$, $b = (\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1}{6})$, with the only nonzero sub-diagonal entries $a_{21} = \frac{1}{2}$, $a_{32} = \frac{1}{2}$, $a_{43} = 1$. For the autonomous system $y' = f(y)$ governing the Robertson problem, one step from y_n at t_n to y_{n+1} at $t_n + h$ reads

$$k_1 = f(y_n), \quad (25)$$

$$k_2 = f(y_n + \frac{h}{2} k_1), \quad (26)$$

$$k_3 = f(y_n + \frac{h}{2} k_2), \quad (27)$$

$$k_4 = f(y_n + h k_3), \quad (28)$$

$$y_{n+1} = y_n + \frac{h}{6} (k_1 + 2k_2 + 2k_3 + k_4). \quad (29)$$

Order conditions up to $p = 3$. We now derive, rather than quote, the conditions on (A, b, c) under which an explicit s -stage RK method reproduces the exact solution's Taylor expansion through $O(h^3)$, and verify that Eq. (24) satisfies them.

For $y' = f(y)$, repeated use of the chain rule gives the exact Taylor expansion of the solution at y_n ,

$$y(t_n + h) = y_n + hf + \frac{h^2}{2}(Df \cdot f) + \frac{h^3}{6} \left[D^2 f(f, f) + Df \cdot Df \cdot f \right] + O(h^4), \quad (30)$$

where $f = f(y_n)$, $Df = Df(y_n)$ is the Jacobian, and $D^2 f$ is the second (bilinear) derivative, all evaluated at y_n .

For a general explicit tableau, write $c_i = \sum_j a_{ij}$ (all stages start from $Y_i(0) = y_n$, so $k_i(0) = f$ for every i) and expand each stage as a power series in h ,

$$k_i(h) = f + h f_i^{(1)} + \frac{h^2}{2} f_i^{(2)} + O(h^3). \quad (31)$$

Since $Y_i(h) = y_n + h \sum_j a_{ij} k_j(h) = y_n + h c_i f + h^2 \sum_j a_{ij} f_j^{(1)} + O(h^3)$, Taylor-expanding $k_i(h) = f(Y_i(h))$ about y_n and matching powers of h gives

$$f_i^{(1)} = c_i (Df \cdot f), \quad f_i^{(2)} = 2 \left(\sum_j a_{ij} c_j \right) (Df \cdot Df \cdot f) + c_i^2 D^2 f(f, f). \quad (32)$$

Substituting into $y_{n+1} = y_n + h \sum_i b_i k_i(h)$ and collecting powers of h ,

$$\begin{aligned} y_{n+1} = y_n &+ h \left(\sum_i b_i \right) f + h^2 \left(\sum_i b_i c_i \right) (Df \cdot f) \\ &+ h^3 \left\{ \left(\sum_i b_i \sum_j a_{ij} c_j \right) (Df \cdot Df \cdot f) + \frac{1}{2} \left(\sum_i b_i c_i^2 \right) D^2 f(f, f) \right\} + O(h^4). \end{aligned} \quad (33)$$

Since f , $Df \cdot f$, $D^2 f(f, f)$, and $Df \cdot Df \cdot f$ are generically linearly independent elementary differentials, matching Eq. (33) to the exact expansion (30) term by term yields the order conditions for $p \leq 3$:

$$\sum_i b_i = 1, \quad (O1)$$

$$\sum_i b_i c_i = \frac{1}{2}, \quad (O2)$$

$$\sum_i b_i c_i^2 = \frac{1}{3}, \quad (O3a)$$

$$\sum_i b_i \sum_j a_{ij} c_j = \frac{1}{6}. \quad (O3b)$$

Condition (O1) matches the h^1 coefficient, (O2) the h^2 coefficient, and (O3a)–(O3b) together match the two independent h^3 terms — the coefficient of $D^2 f(f, f)$ and of $Df \cdot Df \cdot f$ respectively. Together these four scalar equations are necessary and sufficient for order $p = 3$; (O1)–(O2) alone give only order $p = 2$.

Proposition 4 (RK4 satisfies order $p \geq 3$). *The tableau (24) satisfies (O1)–(O3b).*

Proof. With $c = (0, \frac{1}{2}, \frac{1}{2}, 1)$ and $b = (\frac{1}{6}, \frac{1}{3}, \frac{1}{3}, \frac{1}{6})$:

$$\begin{aligned} \sum_i b_i &= \frac{1}{6} + \frac{1}{3} + \frac{1}{3} + \frac{1}{6} = 1, \\ \sum_i b_i c_i &= \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{2} + \frac{1}{6} \cdot 1 = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}, \\ \sum_i b_i c_i^2 &= \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{6} \cdot 1 = \frac{1}{12} + \frac{1}{12} + \frac{2}{12} = \frac{1}{3}. \end{aligned}$$

For (O3b), only stages 3 and 4 contribute a nonzero $\sum_j a_{ij} c_j$ (stage 1 has no predecessors, and $a_{21} c_1 = \frac{1}{2} \cdot 0 = 0$ for stage 2): stage 3 gives $a_{32} c_2 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, stage 4 gives $a_{43} c_3 = 1 \cdot \frac{1}{2} = \frac{1}{2}$. Hence

$$\sum_i b_i \sum_j a_{ij} c_j = b_3 \cdot \frac{1}{4} + b_4 \cdot \frac{1}{2} = \frac{1}{3} \cdot \frac{1}{4} + \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12} + \frac{1}{12} = \frac{1}{6}. \quad \square$$

Order conditions for $p = 4$. A fourth condition set, corresponding to the four rooted trees of order 4, is required for $p = 4$ and is not re-derived here for brevity:

$$\begin{aligned}
\sum_i b_i c_i^3 &= \frac{1}{4}, \\
\sum_i b_i c_i \sum_j a_{ij} c_j &= \frac{1}{8}, \\
\sum_i b_i \sum_j a_{ij} c_j^2 &= \frac{1}{12}, \\
\sum_i b_i \sum_j a_{ij} \sum_k a_{jk} c_k &= \frac{1}{24}.
\end{aligned} \tag{34}$$

(Listed as the standard eight order-4 conditions: the four in (O1)–(O3b) together with these four; see Hairer, Nørsett and Wanner [3], Thm. II.2.1.) Direct substitution of the RK4 tableau into all eight conditions confirms they hold exactly, which is the classical result that Eq. (29) is fourth order; we omit the remaining four substitutions here and cite the standard reference rather than reproduce them, as they add length without new insight beyond Proposition 4.

Application to the Robertson system RK4 is included as the high-accuracy *explicit* control group: it attains fourth-order accuracy at negligible per-step cost whenever the step size respects its stability restriction, unlike the implicit methods of Sections 3.3–3.4 — it has no mechanism for handling stiffness. Its bounded absolute stability region is derived in Section 4.1 and compared against the eigenvalue range of J_r from Section 2.3; Sections 5 and 6.2 show that RK4 is stability-bound as well: its real-axis limit is only about 1.4 times that of Explicit Euler, so its higher order improves the accuracy inside that limit but does not allow larger steps. The observed-vs-theoretical convergence order for RK4 is verified empirically in Section 5.3.

3.3 Implicit Euler Method with Newton-Raphson Iteration

Stage equation and Newton iteration. One step of Implicit Euler, $y_{n+1} = y_n + hf(y_{n+1})$, requires solving the nonlinear stage equation [5]

$$g(Y) := Y - y_n - hf(Y) = 0 \tag{35}$$

for $Y \approx y_{n+1} \in \mathbb{R}^3$. We solve this with Newton–Raphson:

$$Y^{(k+1)} = Y^{(k)} - [Dg(Y^{(k)})]^{-1} g(Y^{(k)}), \quad Dg(Y) = I - hJ(Y), \tag{36}$$

where $J(Y) = Df(Y)$ is the full 3×3 Jacobian of the Robertson vector field (Section 2.1),

$$J(y) = \begin{pmatrix} -0.04 & 10^4 y_3 & 10^4 y_2 \\ 0.04 & -10^4 y_3 - 6 \times 10^7 y_2 & -10^4 y_2 \\ 0 & 6 \times 10^7 y_2 & 0 \end{pmatrix}. \tag{37}$$

The initial guess is the explicit-Euler predictor $Y^{(0)} = y_n + hf(y_n)$. The residual is tested *before* each update, and the iteration is judged converged as soon as $\|g(Y^{(k)})\|_\infty \leq \text{tol}_N$ for a Newton tolerance tol_N set (and swept) in Section 5.5; a step may therefore be accepted with zero Newton updates if the predictor already passes the test. Section 5.5 also tests two alternative initial guesses, the previous solution y_n and a deliberately perturbed predictor.

Relation to the reduced Jacobian of Section 2.3. $J(y)$ above and the 2×2 reduced Jacobian J_r derived in Section 2.3 are *not* the same matrix – they act on spaces of different dimension – but they are two representations of the same underlying linear map, related by the chain rule through the elimination $y_3 = 1 - y_1 - y_2$. Writing $\partial y_3 / \partial y_1 = \partial y_3 / \partial y_2 = -1$,

$$\left. \frac{\partial}{\partial y_i} \right|_{y_3 \text{ elim.}} = \frac{\partial}{\partial y_i} - \frac{\partial}{\partial y_3}, \quad i = 1, 2,$$

which is precisely the tangent-space projection established in Proposition 1 of Section 2.2 ($J(y)$ maps $\mathbb{R}^3$ into $T = \ker(\mathbf{1}^\top)$). Concretely, J_r is obtained from the first two rows of $J(y)$ by subtracting the third column from the first two; we checked numerically that this identity holds to floating-point precision. For a state y on the invariant plane $\mathcal{M} = \{\mathbf{1}^\top y = 1\}$ this gives $\text{spec } J(y) = \{0\} \cup \text{spec } J_r(y_1, y_2)$ (Proposition 3 of Section 2.2), and hence

$$\text{spec } Dg(y) = \{1\} \cup \{1 - h\lambda_{\text{slow}}, 1 - h\lambda_{\text{fast}}\}, \quad (38)$$

where $\lambda_{\text{slow}}, \lambda_{\text{fast}}$ are the eigenvalues of J_r analysed in Section 4.2. The 2×2 form thus drives the eigenvalue/stiffness analysis, while the full 3×3 form drives the Newton solve here; the extra zero eigenvalue of J only contributes the harmless eigenvalue 1 to Dg .

Proposition 5 (exact identity for the linear invariant). Let $\hat{Y} \in \mathbb{R}^3$ be arbitrary – in particular any Newton iterate, converged or not – and define its residual $r := g(\hat{Y}) = \hat{Y} - y_n - hf(\hat{Y})$. Then

$$\mathbf{1}^\top \hat{Y} = \mathbf{1}^\top y_n + \mathbf{1}^\top r$$

holds as an exact algebraic identity, not an approximation.

Proof. By definition of r ,

$$\mathbf{1}^\top r = \mathbf{1}^\top \hat{Y} - \mathbf{1}^\top y_n - h \mathbf{1}^\top f(\hat{Y}).$$

Proposition 1 of Section 2.2 established $\mathbf{1}^\top f(y) \equiv 0$ for *every* $y \in \mathbb{R}^3$, not merely for y on the trajectory or satisfying the stage equation; in particular it holds at $y = \hat{Y}$ regardless of how close $\hat{Y}$ is to the true root. Hence the last term vanishes and rearranging gives the claim. $\square$

The identity says that the conservation defect of an iterate *equals* $\mathbf{1}^\top r$. It does not say that this defect is large when r is large: for Newton’s method with the analytic Jacobian the defect is in fact zero after every update, as we now show.

Proposition 6 (Newton updates preserve the linear invariant). Let $\tilde{J} \in \mathbb{R}^{3 \times 3}$ be any matrix with $\mathbf{1}^\top \tilde{J} = 0$ (for example $\tilde{J} = J(Y)$ at any Y , or a frozen Jacobian), and let $Y^+ = Y - (I - h\tilde{J})^{-1}g(Y)$. Then $\mathbf{1}^\top Y^+ = \mathbf{1}^\top y_n$ for every $Y \in \mathbb{R}^3$. Consequently, if the initial guess satisfies $\mathbf{1}^\top Y^{(0)} = \mathbf{1}^\top y_n$, then every Newton iterate does, and $\mathbf{1}^\top r^{(k)} = 0$ for all k .

Proof. Since $\mathbf{1}^\top \tilde{J} = 0$, we have $\mathbf{1}^\top (I - h\tilde{J}) = \mathbf{1}^\top$, hence $\mathbf{1}^\top (I - h\tilde{J})^{-1} = \mathbf{1}^\top$. Therefore $\mathbf{1}^\top Y^+ = \mathbf{1}^\top Y - \mathbf{1}^\top g(Y)$, and by the Proposition above $\mathbf{1}^\top g(Y) = \mathbf{1}^\top Y - \mathbf{1}^\top y_n$, which gives $\mathbf{1}^\top Y^+ = \mathbf{1}^\top y_n$. The explicit-Euler predictor and the previous solution both satisfy $\mathbf{1}^\top Y^{(0)} = \mathbf{1}^\top y_n$ (the first because $\mathbf{1}^\top f \equiv 0$), and so does any perturbation of them with zero component sum. Finally $\mathbf{1}^\top r^{(k)} = \mathbf{1}^\top Y^{(k)} - \mathbf{1}^\top y_n = 0$. $\square$

Corollary (conservation defect of the implicit solvers). (i) With the analytic Jacobian and an initial guess as above, the single-step defect $\mathbf{1}^\top y_{n+1} - \mathbf{1}^\top y_n$ is zero in exact arithmetic *for every* tol_N , and of the order of rounding error in floating point. It is therefore *not* a diagnostic of Newton convergence, and a small conservation defect must not be read as evidence that the stage equation has been solved. (ii) If $\mathbf{1}^\top \tilde{J} \neq 0$ (e.g. an inaccurate finite-difference Jacobian) or

the initial guess has a non-zero component sum, no such cancellation occurs, but the Proposition above still bounds the defect by the stopping test: by Hölder's inequality

$$|\mathbf{1}^\top \hat{Y} - \mathbf{1}^\top y_n| = |\mathbf{1}^\top r| \leq \|\mathbf{1}\|_1 \|r\|_\infty \leq 3 \text{tol}_N,$$

and, writing $\delta_n := \mathbf{1}^\top y_n - 1$, the defects accumulate as $\delta_{n+1} = \delta_n + \mathbf{1}^\top r_{n+1}$, so that only $|\delta_N| \leq 3N \text{tol}_N$ can be guaranteed after N steps. This is a worst-case bound, not a prediction. The Newton-tolerance study of Section 5.5 is consistent with (i): the defect stays at rounding level for all tolerances swept, including those for which the computed solution is non-physical.

Numerical confirmation. We checked these statements on the Robertson vector field (not just the test equation), with $h = 10^{-2}$ and $Y^{(0)} = y_n = (1, 0, 0)$, deliberately *not* running Newton to convergence. After one update, $\|g\|_\infty \approx 4.8 \times 10^{-2}$ ($\|g\|_2 \approx 6.8 \times 10^{-2}$); after two, $\|g\|_\infty \approx 1.2 \times 10^{-2}$ ($\|g\|_2 \approx 1.7 \times 10^{-2}$). In both cases $\mathbf{1}^\top \hat{Y} - \mathbf{1}^\top y_n$ vanished to machine precision and $|\mathbf{1}^\top r| \lesssim 3 \times 10^{-17}$, i.e. the identity of the first Proposition holds and $\mathbf{1}^\top r \approx 0$ as the second predicts. As a contrast, replacing J by a Jacobian with a 0.1% error in the (2, 2) entry (so that $\mathbf{1}^\top \tilde{J} \neq 0$) gave, after two updates, $\|g\|_\infty \approx 1.2 \times 10^{-2}$ and a defect of about 5×10^{-5} , non-zero but well inside the bound $3\|r\|_\infty$ of part (ii).

Local convergence of Newton's method. If f is continuously differentiable with a Lipschitz continuous Jacobian near the exact root Y^* and $Dg(Y^*)$ is nonsingular, then for $Y^{(k)}$ sufficiently close to Y^*

$$\|Y^{(k+1)} - Y^*\| \leq C \|Y^{(k)} - Y^*\|^2$$

for some constant C , which is the standard quadratic convergence result for Newton iteration [6, 7]. Here $Dg(Y^*)$ is nonsingular whenever $1/h$ is not an eigenvalue of $J(Y^*)$; on $\mathcal{M}$ this is guaranteed for all $h > 0$ by (38), because the relevant eigenvalues of J_r have non-positive real part along the trajectory.

One problem-specific caveat is worth flagging here rather than only in the results. Along the reference trajectory the eigenvalues of J are bounded by $|\lambda|_{\max} \approx 3.4 \times 10^3$ (Section 4.2), *not* by the rate constant 3×10^7 ; by (38) the eigenvalues of Dg lie in $[1, 1 + h|\lambda|_{\max}]$. At the reference states sampled on our 200-point output grid, $\text{cond}_2(I - hJ)$ is at most about 6 for $h = 10^{-3}$ and about 59 for $h = 10^{-2}$, so the Newton matrix is well conditioned there. Large eigenvalues of order $6 \times 10^7 y_2$ (from the entry $2k_2 y_2$ of J) appear only when an iterate has a large y_2 ; e.g. at $y = (0.5, 0.1, 0.1)$ and $h = 10^{-2}$ the condition number is about 1.4×10^5 . Such states lie far from the trajectory, where $y_2 \approx 10^{-5}$, and where the basin of quadratic convergence may shrink and damped or globalised Newton steps might be needed. Whether this matters in practice is quantified by the iteration-count and tolerance sweep of Section 5.5 rather than asserted here.

3.4 Trapezoidal Method (Crank–Nicolson) as a Second Implicit Scheme

To complement the first-order Implicit Euler scheme and establish a multi-order comparison framework, we introduce the classical Trapezoidal method as a second implicit integrator [5]. The method discretizes the autonomous system $\frac{dy}{dt} = f(y)$ by averaging the vector fields at the current and next time steps:

$$y_{n+1} = y_n + \frac{h}{2} [f(y_n) + f(y_{n+1})], \quad (39)$$

where $h > 0$ denotes the integration step size.

Non-linear Residual and Newton–Raphson Formulation Following the iterative framework established in Section 3.3, Eq. (39) defines an implicit algebraic system for the unknown

state vector y_{n+1} . We recast this relation into finding the root of the non-linear residual vector function $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$:

$$G(y_{n+1}) \equiv y_{n+1} - y_n - \frac{h}{2}f(y_n) - \frac{h}{2}f(y_{n+1}) = 0. \quad (40)$$

Solving $G(y_{n+1}) = 0$ via the multivariate Newton–Raphson method yields the linear update step at iteration k :

$$W \left(y_{n+1}^{(k)} \right) \Delta y^{(k)} = -G \left(y_{n+1}^{(k)} \right), \quad y_{n+1}^{(k+1)} = y_{n+1}^{(k)} + \Delta y^{(k)}, \quad (41)$$

where the iteration matrix $W(y) \in \mathbb{R}^{d \times d}$ corresponds to the Fréchet derivative $DG(y)$:

$$W(y) \equiv DG(y) = \frac{\partial G}{\partial y}(y) = I - \frac{h}{2}J_f(y). \quad (42)$$

Here $I \in \mathbb{R}^{d \times d}$ is the identity matrix and $J_f(y) \equiv \partial f / \partial y(y)$ is the full 3×3 Jacobian of Eq. (37); its reduced 2×2 counterpart J_r (Section 2.3) is used for the spectral analysis only. The conservation results of Section 3.3 carry over: since $\mathbf{1}^T f \equiv 0$ one has $\mathbf{1}^T G(Y) = \mathbf{1}^T Y - \mathbf{1}^T y_n$, and since $\mathbf{1}^T J_f \equiv 0$ every Newton update with matrix $W = I - \frac{h}{2}J_f$ maps Y to a point with $\mathbf{1}^T Y^+ = \mathbf{1}^T y_n$.

Second-Order Accuracy By symmetric Taylor expansion of the stage evaluations around $t_n + \frac{h}{2}$, the local truncation error (LTE) of the Trapezoidal method satisfies:

$$\tau_{n+1} = y(t_{n+1}) - y_{n+1} = \mathcal{O}(h^3). \quad (43)$$

Accumulating LTE over $N = T/h$ integration steps yields a *second-order global discretization error*:

$$E_{\text{global}}(h) = \max_{0 \leq n \leq N} \|y(t_n) - y_n\| = \mathcal{O}(h^2). \quad (44)$$

This offers a higher order of accuracy ($\mathcal{O}(h^2)$) compared to the first-order ($\mathcal{O}(h)$) convergence of the Implicit Euler scheme.

Role in Multi-Order Cross-Consistency Verification The inclusion of the second-order Trapezoidal scheme serves an essential structural role in our overall numerical validation strategy. Together with the first-order Implicit Euler method, it forms the foundation for evaluating numerical self-consistency across methods of different orders—representing the third component of our threefold validation protocol detailed in Section 5.3.

Remark on Asymptotic Behavior: While the Trapezoidal scheme achieves second-order accuracy, its asymptotic damping characteristics under severe stiffness differ fundamentally from Implicit Euler. The rigorous stability analysis regarding its lack of L -stability (despite being A -stable) is detailed in Section 4.1.

3.5 Adaptive Step-Size Control

Step doubling is used as a common local-error controller for both Explicit Euler (EE) and Implicit Euler (IE), following standard adaptive time-stepping ideas [8, 9]. The controller is separated from the base method: changing the step size alone does not remove an explicit method’s stability restriction.

3.5.1 Common step-doubling controller

Let Φ_h denote one step of a first-order base method. Starting at $\mathbf{y}_n$, compute

$$\begin{aligned} \mathbf{y}_{\text{full}} &= \Phi_h(\mathbf{y}_n), & \mathbf{y}_{\text{mid}} &= \Phi_{h/2}(\mathbf{y}_n), \\ \mathbf{y}_{\text{two}} &= \Phi_{h/2}(\mathbf{y}_{\text{mid}}), & e_n &= \|\mathbf{y}_{\text{two}} - \mathbf{y}_{\text{full}}\|_{\infty}. \end{aligned} \quad (45)$$

The leading local errors are Ch^2 and $Ch^2/2$, respectively. Thus the Richardson divisor $2^p - 1$ equals one for $p = 1$, and e_n estimates the leading local error of the two-half-step value. The trial is normally accepted when $e_n \leq \text{tol}$; the step-size update is

$$h_{\text{new}} = h \, 0.9 \left(\frac{\text{tol}}{e_n} \right)^{1/2}. \quad (46)$$

The multiplier is clipped to $[0.5, 2]$ after acceptance and $[0.1, 0.5]$ after rejection, with a factor of two for zero estimated error. Both methods start at $h_0 = 10^{-4}$ and use $h_{\min} = 10^{-10}$. The explicit method retains $h_{\max} = 10^{-2}$; the implicit method uses $h_{\max} = 5$, allowing accuracy to choose substantially larger steps. Hence their step-size ceilings are deliberately different and are reported with the results. The norm test is absolute, so it does not give relative control of the small component y_2 .

The original explicit implementation retains a minimum-step acceptance fallback at $h \leq 1.01h_{\min}$; this is not used in the completed runs reported here. The new implicit implementation never accepts a step with $e_n > \text{tol}$: failure at the minimum step, or inability to advance time, terminates with an explicit error.

3.5.2 Adaptive Explicit Euler

For EE, $\Phi_h(\mathbf{y}) = \mathbf{y} + h\mathbf{f}(\mathbf{y})$ and the accepted state is $\mathbf{y}_{n+1} = \mathbf{y}_{\text{two}}$. This remains a first-order method. Its retained experiment is described below.

Demonstration run ($\text{tol} = 10^{-6}$). Over $0 \leq t \leq 40$ the controller accepts 25 398 steps and rejects 12 686 trial steps (33% of all trials). The accepted step sizes range from 1.0×10^{-4} to 8.3×10^{-3} , with mean 1.57×10^{-3} and median 1.73×10^{-3} , so the step is far from uniform. In the initial transient it grows quickly: after 11 accepted steps ($t \approx 7 \times 10^{-3}$) it exceeds 3×10^{-3} . It does *not*, however, keep growing where the solution is smooth: the median step is 1.9×10^{-3} for $1 \leq t < 10$ and only 1.4×10^{-3} for $t \geq 10$, decreasing to about 9×10^{-4} near $t = 40$, although y_2 then varies slowly (Figure 1a).

The final state is

$$\mathbf{y}(40) \approx (0.715813151, 9.14185 \times 10^{-6}, 0.284177707),$$

against the reference $\mathbf{y}_{\text{ref}}(40) \approx (0.715827069, 9.1855 \times 10^{-6}, 0.284163746)$ of Section 5.1. The final-state error is $\|\mathbf{y}(40) - \mathbf{y}_{\text{ref}}(40)\|_2 \approx 1.97 \times 10^{-5}$; the maximum over the whole trajectory is $E_{\infty} = 1.97 \times 10^{-5}$ and is attained at the end. The invariant defect is 1.33×10^{-14} and no negative component occurs.

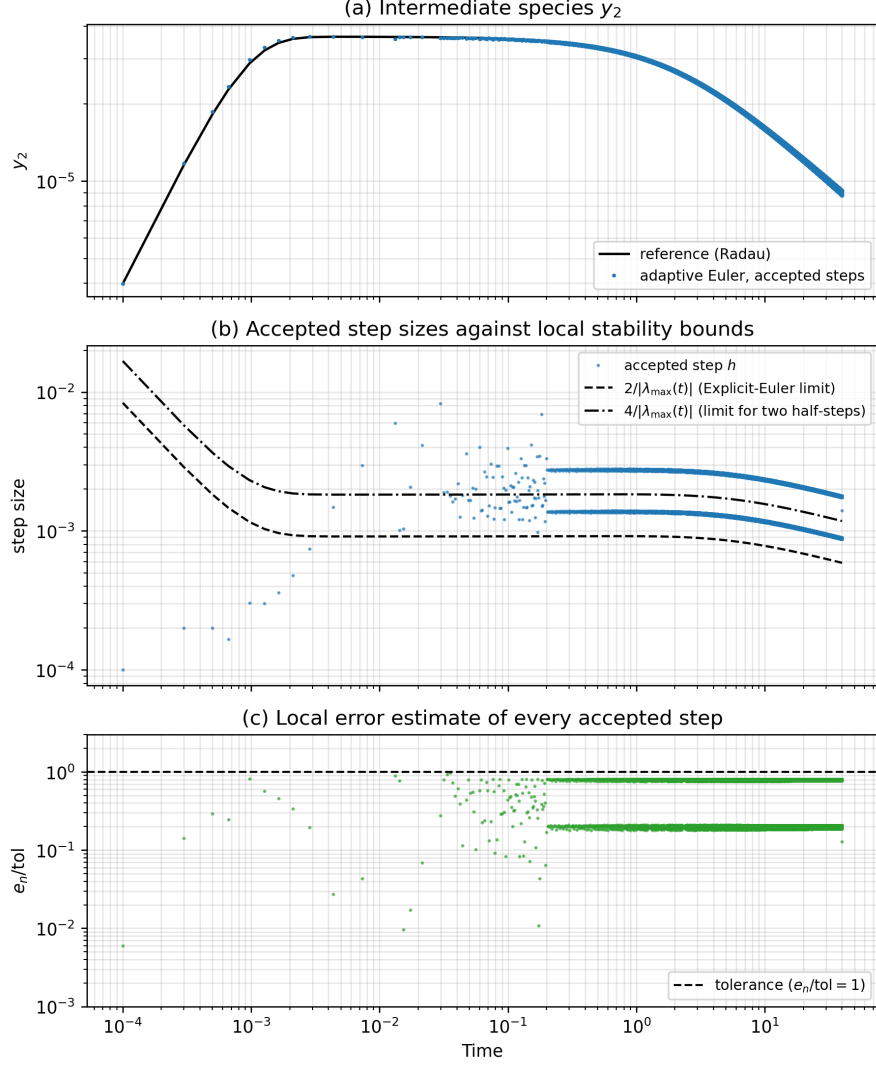


Figure 1: Adaptive Explicit Euler, $\text{tol} = 10^{-6}$, every accepted step shown. (a) y_2 against the reference. (b) Accepted step sizes against the local stability bounds $2/|\lambda_{\max}(t)|$ (Explicit Euler) and $4/|\lambda_{\max}(t)|$ (two half-steps), with $|\lambda_{\max}|$ evaluated on the reference trajectory (Section 4.2). (c) Local error estimate e_n of every accepted step in units of the tolerance.

Local versus global error. Panel (c) of Figure 1 shows that the controller does what it is designed to do locally: the estimate of every accepted step lies below the tolerance (maximum 0.97 tol), and for 90% of the steps between 0.19 and 0.81 of it, i.e. within a factor of about five. The global error is nevertheless about 20 times the tolerance. The tolerance bounds a per-step estimate, whereas the global error accumulates over 25 398 steps; the sum of the local estimates is 1.3×10^{-2} (at most $N \text{ tol} = 2.5 \times 10^{-2}$) and is much larger than the actual global error, which indicates that the stiff dynamics damp most of the local errors, but that the global error is not controlled by tol .

Why the step does not grow. Panel (b) explains the behaviour of h . For $t \gtrsim 0.2$ the accepted steps alternate between two levels: a lower one below $4/|\lambda_{\max}|$, whose two half-steps of size $h/2$ lie inside the Explicit-Euler stability interval, and an upper one above it, with local error estimates of about 0.2 and 0.8 of the tolerance (panel c). For $t > 0.1$ the accepted steps lie between 1.5 and 3.0 times the Explicit-Euler limit $2/|\lambda_{\max}(t)|$ (5th–95th percentile). The controller therefore rides the linear stability boundary of the explicit method and stays

there only through rejections, which are about one third of all trial steps. Adaptive step-size control is thus an efficiency improvement for an explicit method, not a way around stiffness; its dependence on the tolerance is quantified in Section 5.5.2, and Section 5.6 compares it with an adaptive implicit solver.

3.5.3 Adaptive Implicit Euler and local extrapolation

For IE, each full or half step solves

$$\mathbf{z} - \mathbf{y} - \Delta t \mathbf{f}(\mathbf{z}) = \mathbf{0}, \quad (I - \Delta t J(\mathbf{z}^{(k)})) \delta^{(k)} = -F(\mathbf{z}^{(k)}).$$

The implementation reuses the analytic Jacobian and full Newton formulation of Section 3.3, with an explicit-Euler predictor and an absolute residual tolerance $\tau_N = 10^{-13}$. A failed or non-finite solve rejects the trial and reduces h by a factor of four. Newton updates include the work spent on rejected trials. Two accepted-state choices are tested:

$$\mathbf{y}_{n+1} = \mathbf{y}_{\text{two}} \quad (\text{first order}), \quad \mathbf{y}_{n+1} = 2\mathbf{y}_{\text{two}} - \mathbf{y}_{\text{full}} \quad (\text{local extrapolation}). \quad (47)$$

The extrapolation cancels the leading $O(h^2)$ local term, giving formal second order for smooth solutions with sufficiently accurate nonlinear solves. The controller still uses the $O(h^2)$ difference in Eq. (45), so its exponent remains 1/2, not 1/3. For the extrapolated state the estimator is a lower-order, typically conservative proxy, not an estimate of its own $O(h^3)$ local error and not a rigorous upper bound in every stiff regime.

Linear invariant. Each exactly solved IE substep preserves $\mathbf{1}^T \mathbf{y}$, since $\mathbf{1}^T \mathbf{f} = 0$. Therefore, if $\mathbf{1}^T \mathbf{y}_{\text{full}} = \mathbf{1}^T \mathbf{y}_{\text{two}} = m_n$, then

$$\mathbf{1}^T (2\mathbf{y}_{\text{two}} - \mathbf{y}_{\text{full}}) = 2m_n - m_n = m_n.$$

With Newton solves the identity holds up to algebraic and floating-point error; $\mathbf{1}^T J = 0$ explains why a Newton update restores the invariant. This affine-combination argument does not guarantee non-negativity, which is checked independently.

Implicit demonstration at $\text{tol} = 1 \times 10^{-6}$. The extrapolated variant accepts 501 steps, rejects 0 trials and performs 2029 Newton updates. Its global error is $E_\infty = 9.00 \times 10^{-7}$ and its invariant defect is 2.4×10^{-15} , with no negative concentration in this run. For $t > 0.1$, the 5th–95th percentiles of $h/(2/|\lambda_{\max}|)$ are 18–348, with median 74. The accepted implicit steps thus pass far beyond the explicit stability bound. At the same nominal absolute tolerance, this run uses about 51 times fewer accepted steps and has about 22 times smaller global error than adaptive EE. Equal nominal tolerances are not equal-accuracy comparisons; the measured work–precision study is in Section 5.6.

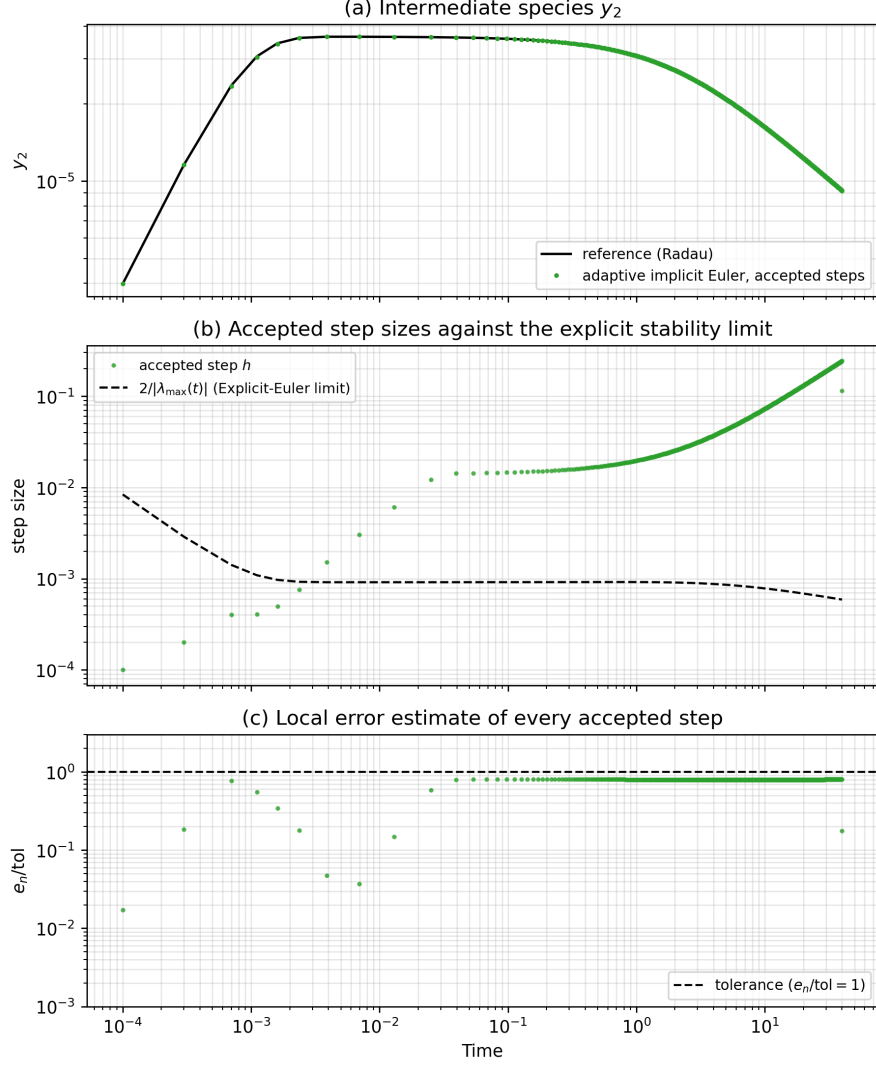


Figure 2: Self-implemented adaptive Implicit Euler with local extrapolation, $\text{tol} = 10^{-6}$. (a) Intermediate species against Radau. (b) Accepted macro-step sizes against the explicit stability bound. (c) The first-order step-doubling proxy in units of tolerance; it is not the local error of the extrapolated state.

4 Linear Stability Analysis

This section collects the local (linear) stability tools and applies them to the Robertson system. Section 4.1 derives the absolute-stability regions of the four integrators for the scalar test equation; Section 4.2 computes the Jacobian eigenvalues, the stiffness ratio and a non-normality diagnostic along our own reference trajectory; Section 4.3 turns these into a step-size prediction for the explicit methods and compares it with the experiments of Section 5.4. Throughout, “local” means that the Jacobian is frozen at a state of the reference trajectory.

4.1 Absolute Stability Regions of the Four Methods

To characterise each integrator independently of the nonlinear system, we apply it to the scalar test equation $y' = \lambda y$, $\lambda \in \mathbb{C}$, and set $z = h\lambda$. Standard absolute-stability theory is used here [10, 5]. One step of any of the four methods of Section 3 reads $y_{n+1} = R(z)y_n$ with a method-specific growth factor $R(z)$, and the region of absolute stability is $\mathcal{S} = \{z \in \mathbb{C} : |R(z)| \leq 1\}$.

Explicit Euler. $y_{n+1} = (1 + z)y_n$, so $R(z) = 1 + z$. The region $|1 + z| \leq 1$ is the closed unit disk centred at $z = -1$; on the real axis it gives $z \in [-2, 0]$.

RK4. For an explicit Runge–Kutta method of order p applied to the test equation, $R(z)$ is the degree- p Taylor polynomial of e^z ; for classical RK4,

$$R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}.$$

On the negative real axis R is positive and not monotone: it decreases from $R(0) = 1$ to a minimum $R \approx 0.270$ at $z \approx -1.60$ and then increases again, returning to $R = 1$ at $z \approx -2.7853$ (root of $R(x) = 1$ found with Brent’s method). The real-axis stability interval is therefore $[-2.7853, 0]$, about 1.4 times that of Explicit Euler but still bounded. Away from the real axis the region has the bulging shape of Figure 3 (top right).

Implicit Euler. $y_{n+1} = y_n + z y_{n+1}$, hence $R(z) = 1/(1 - z)$. *A-stability.* For $z = x + iy$ with $x \leq 0$, $|1 - z|^2 = (1 - x)^2 + y^2 = 1 - 2x + x^2 + y^2 \geq 1$, so $|R(z)| \leq 1$ on the closed left half-plane. In fact $\mathcal{S} = \{z : |z - 1| \geq 1\}$ is strictly larger: only the open unit disk centred at $z = 1$ is excluded (Figure 3, bottom left). *L-stability.* $R(z) \rightarrow 0$ as $z \rightarrow -\infty$ (numerically $R(-10^8) \approx 1.0 \times 10^{-8}$), so the fastest decaying modes are damped strongly: Implicit Euler is A-stable and L-stable.

Trapezoidal / Crank–Nicolson. $y_{n+1} = y_n + \frac{h}{2}\lambda(y_n + y_{n+1})$, hence $R(z) = \frac{1 + z/2}{1 - z/2}$. *A-stability, exactly.* With $w = z/2 = u + iv$, $|1 + w| \leq |1 - w| \iff (1 + u)^2 \leq (1 - u)^2 \iff u \leq 0$. The stability region is exactly the closed left half-plane, and on the imaginary axis $|R(iy)| = 1$ (numerically $||R(iy)| - 1| \leq 4 \times 10^{-16}$ for $10^{-3} \leq y \leq 10^3$). *Not L-stable.* $R(z) \rightarrow -1$ as $z \rightarrow -\infty$ (numerically $R(-10^8) \approx -1.0000$). Very fast decaying modes are therefore reflected with amplitude close to one and alternating sign instead of being damped. This is a concrete, testable prediction; Section 5.6 confirms that Trapezoidal/CN loses non-negativity at large steps ($h = 5 \times 10^{-3}$ and 10^{-2}) although it is A-stable.

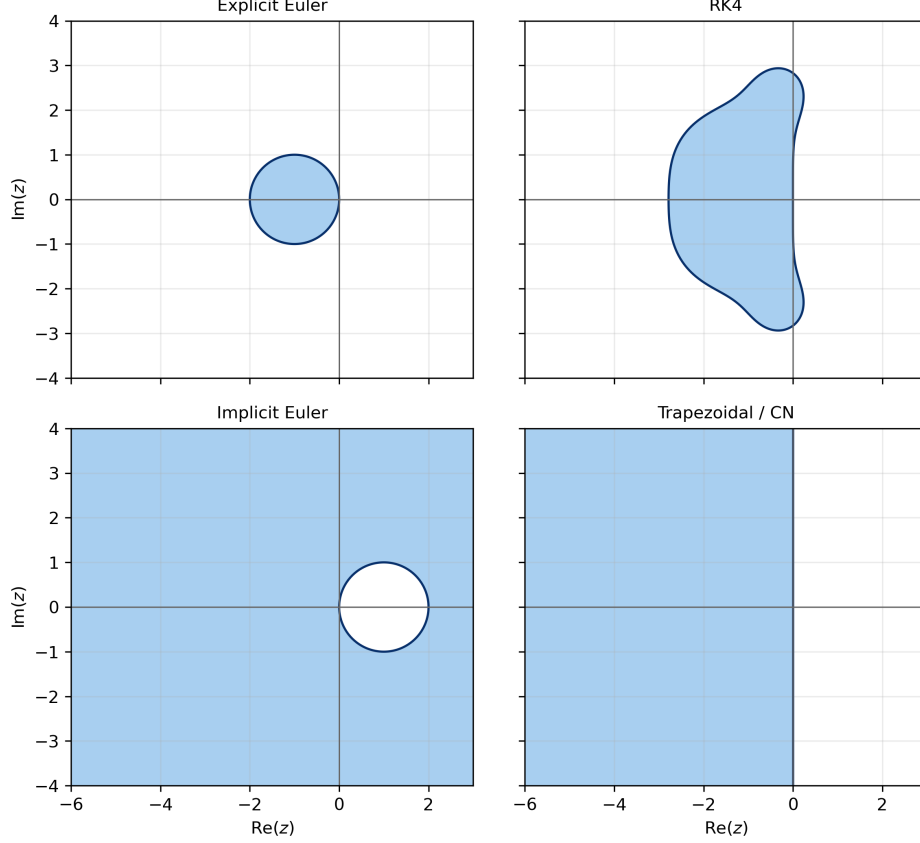


Figure 3: Absolute-stability regions $|R(z)| \leq 1$ (shaded) of the four integrators on the common domain $\text{Re } z \in [-6, 3]$, $\text{Im } z \in [-4, 4]$.

Caveat. These regions describe the scalar, linear, autonomous test equation only; they say nothing directly about the nonlinear Robertson system. A nonlinear theory exists in the form of Dahlquist’s B-stability [5], which guarantees $|y_{n+1} - z_{n+1}| \leq |y_n - z_n|$ under the one-sided Lipschitz condition $\langle f(u) - f(v), u - v \rangle \leq 0$, but the condition is not established for the Robertson vector field, which contains indefinite quadratic terms. A-/L-stability are therefore used as necessary local diagnostics, to be corroborated by the eigenvalue analysis of Section 4.2 and the step-size sweep of Section 5.4.

4.2 Eigenvalue Analysis of the Robertson System

Setup. The Jacobian is evaluated along our own reference trajectory (Radau, $\text{rtol} = 10^{-12}$, $\text{atol} = 10^{-14}$, Section 5.1) through its dense output, on the 200 positive output times (log-spaced points in $[10^{-8}, 40]$; $t = 0$ is excluded because the reduced Jacobian is degenerate there), for smooth curves, on a 1000-point logarithmic grid on the same interval.

Structural zero and initial degeneracy. Because $\mathbf{1}^T \mathbf{f} \equiv 0$, the full 3×3 Jacobian satisfies $\mathbf{1}^T J(\mathbf{y}) = \mathbf{0}^T$ for every state: it has the eigenvalue 0 for all t (left eigenvector $\mathbf{1}$), and a stiffness ratio that includes it is infinite or undefined. Eliminating $y_3 = 1 - y_1 - y_2$ gives the reduced Jacobian $J_r = [J_{ij} - J_{i3}]_{i,j=1,2}$ of Section 2.3. The plane $\mathbf{1}^T \mathbf{v} = 0$ is invariant under J , so the two eigenvalues of J_r are exactly the two nonzero eigenvalues of J ; we confirmed this numerically (largest relative difference 6×10^{-15} over all sampled states). We therefore define

$$S(t) = \frac{\max\{|\lambda_{\text{slow}}|, |\lambda_{\text{fast}}|\}}{\min\{|\lambda_{\text{slow}}|, |\lambda_{\text{fast}}|\}}$$

from the eigenvalues of J_r . If the full Jacobian is used instead, an eigenvalue is classified as zero when $|\lambda| < 10^{-9}$: along the trajectory the structural zero has modulus at most 1×10^{-15} , while the smallest nonzero eigenvalue is 0.0214 (at $t = 40$), so the threshold is unambiguous.

The initial state $(1, 0, 0)$ is a *different*, genuine degeneracy: with $y_2 = y_3 = 0$ the fast reaction terms vanish, $J(0)$ has a double zero eigenvalue, and $J_r(0) = \begin{pmatrix} -0.04 & 0 \\ 0.04 & 0 \end{pmatrix}$ has a zero eigenvalue in addition to -0.04 . The ratio $S(0)$ is therefore undefined and all curves start at the first positive output time $t = 10^{-8}$, where S is finite ($S \approx 1.67$).

Results. Figure 4 shows $|\lambda_{\text{slow}}|$, $|\lambda_{\text{fast}}|$ and $S(t)$. Two features deserve comment. (i) For small t the growing eigenvalue is $\lambda \approx -2k_2y_2 \approx -2k_2k_1t$, while the other stays near $-k_1 = -0.04$. The two moduli coincide at $t_c = 1/(2k_2) \approx 1.67 \times 10^{-8}$; for $t < t_c$ the ordering by modulus is reversed, so $S = \max/\min$ touches 1 at t_c (a property of the definition, not of the dynamics). (ii) For $t \gtrsim 10^{-3}$, $|\lambda_{\text{fast}}|$ saturates near 2.2×10^3 and later grows to 3.393×10^3 at $t = 40$ as y_3 increases, whereas $|\lambda_{\text{slow}}| \approx 0.4$ decays to 0.0214; hence S has a plateau near 5.4×10^3 and rises to 1.58×10^5 at $t = 40$.

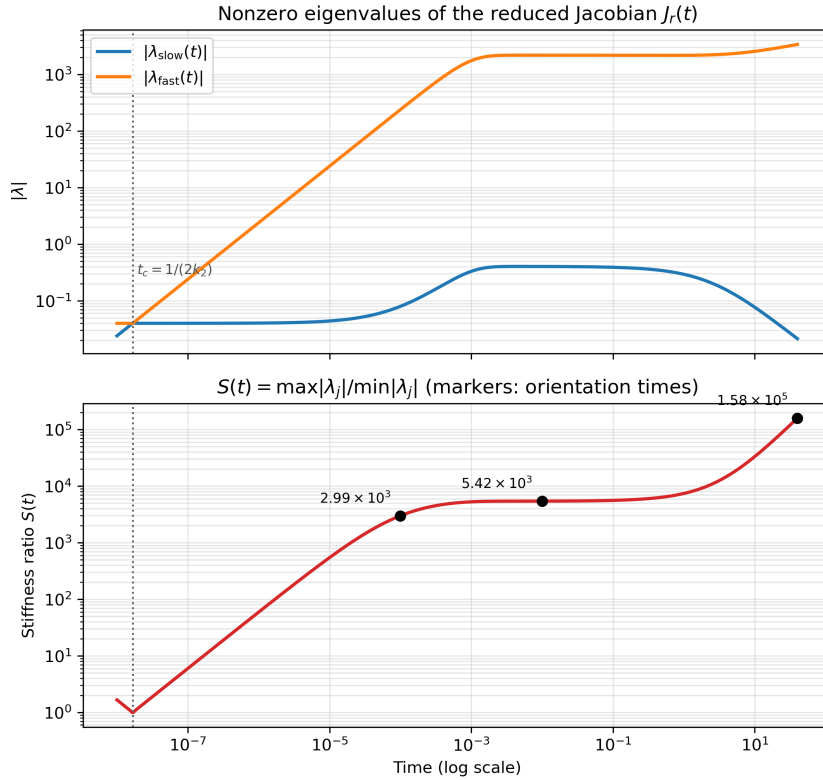


Figure 4: Top: magnitudes of the two nonzero eigenvalues of $J_r(t)$ along the reference trajectory. Bottom: stiffness ratio $S(t)$, starting at the first positive output time; markers are the orientation times of Table 1. The dotted line marks $t_c = 1/(2k_2)$.

Table 1: Stiffness ratio $S(t)$ from the reduced Jacobian along our own reference, evaluated at the exact orientation times, compared with the Problem-Pack check values (which are quoted to two or three digits and used only as a self-consistency target).

t	our $S(t)$	Problem-Pack value	relative difference
1×10^{-4}	2.99×10^3	3.0×10^3	0.23%
1×10^{-2}	5.42×10^3	5.4×10^3	0.42%
40	1.58×10^5	1.58×10^5	0.25%

Table 1 compares $S(t)$ at the three orientation times (evaluated exactly, not at the nearest grid point) with the Problem-Pack check values; the differences are below 0.5%, i.e. within the two to three digits in which those values are quoted. At $t = 40$, $|\lambda|_{\max} = 3.393 \times 10^3$ and

$$J_r(40) \approx \begin{pmatrix} -0.1319 & 2841.5 \\ 0.1319 & -3392.7 \end{pmatrix}.$$

Non-normality. J_r is not symmetric, so eigenvalues alone can mislead. For a diagonalisable $J = V\Lambda V^{-1}$, one Explicit-Euler step matrix satisfies

$$\|(I + hJ)^n\|_2 \leq \kappa(V) \max_j |1 + h\lambda_j|^n, \quad \kappa(V) = \|V\|_2 \|V^{-1}\|_2,$$

so $\kappa(V)$ bounds how much the eigenvalue-based factor can understate transient amplification ($\kappa = 1$ for a normal matrix). Figure 5 shows $\kappa(V(t))$: on the 200 output times its median is 1.014, only 2% of the points exceed 10, and $\kappa = 2.14$ at $t = 40$. The peak (54.3 at $t = 1.74 \times 10^{-8}$ on the output grid) is a resolution artefact: it sits at the eigenvalue crossing t_c , where the two eigenvalues coincide and J_r is nearly a Jordan block, so the peak grows as the grid is refined (229 on the 1000-point grid). Where it occurs $|\lambda| \approx 0.04$, so $h|\lambda| \ll 2$ for every step size used and non-normality cannot influence stability. At late times, where $h|\lambda_{\text{fast}}| \approx 2$ decides stability, $\kappa \lesssim 2.2$, so the frozen-Jacobian estimate of the next section is reliable to within that factor.

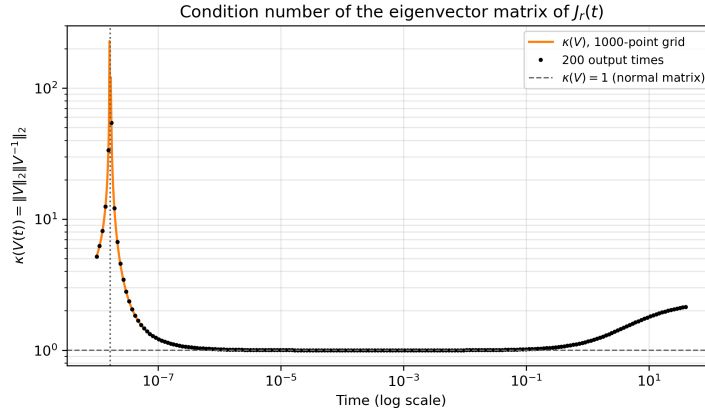


Figure 5: Condition number $\kappa(V(t))$ of the eigenvector matrix of $J_r(t)$. Dots: the 200 output times; line: 1000-point grid. The dotted line marks t_c .

4.3 Step-Size Constraints: Theory versus Experiment

Prediction. For a real negative eigenvalue the Explicit-Euler condition $|1 + h\lambda| \leq 1$ gives $h \leq 2/|\lambda|$. Freezing the Jacobian at the state $\mathbf{y}_{\text{ref}}(t)$, the local restriction is $h \lesssim 2/|\lambda_{\text{fast}}(t)|$, and a single step size valid on all of $[0, 40]$ must satisfy

$$h < \frac{2}{\max_{t \in [0, 40]} |\lambda_{\text{fast}}(t)|} = \frac{2}{|\lambda|_{\max}} = \frac{2}{3.393 \times 10^3} \approx 5.895 \times 10^{-4}, \quad (48)$$

the maximum being attained at $t = 40$. The same argument with the RK4 real-axis limit gives $h < 2.7853/|\lambda|_{\max} \approx 8.21 \times 10^{-4}$. Implicit Euler and Trapezoidal/CN are A-stable, so linear stability imposes no restriction on h ; accuracy and admissibility still have to be checked (Section 5.4).

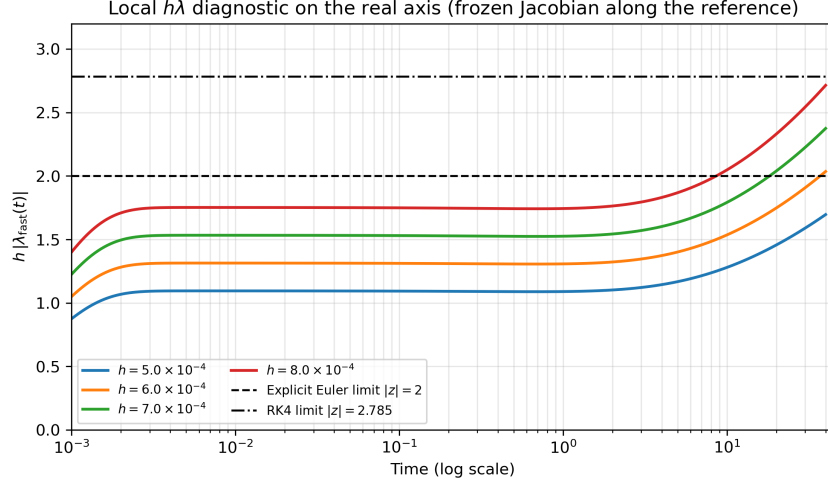


Figure 6: Local $h\lambda$ diagnostic on the real axis: $h|\lambda_{\text{fast}}(t)|$ along the reference trajectory for four step sizes, against the real-axis limits of Explicit Euler (2) and RK4 (2.7853). A curve above a limit means that the frozen-Jacobian condition is violated at that time.

Figure 6 shows where the condition fails. For $h = 5 \times 10^{-4}$ it holds throughout; for $h = 6 \times 10^{-4}$, 7×10^{-4} and 8×10^{-4} it is first violated at $t \approx 37.0$, 18.2 and 8.6. The RK4 step $h = 8 \times 10^{-4}$ used in the convergence study reaches only 97% of its limit at $t = 40$ and stays below it.

Table 2: Explicit Euler: time at which the local limit $h|\lambda_{\text{fast}}(t)| > 2$ is first exceeded along the reference trajectory, compared with the outcome of the step-size sweep of Section 5.4.

h	local limit exceeded from $t =$	observed outcome	divergence time
4.00×10^{-4}	never	admissible	—
5.00×10^{-4}	never	admissible	—
5.50×10^{-4}	never	admissible	—
5.90×10^{-4}	39.9	admissible	—
5.95×10^{-4}	38.4	admissible	—
6.00×10^{-4}	37.0	reaches T , negative values	—
7.00×10^{-4}	18.2	diverged	27.1
8.00×10^{-4}	8.6	diverged	17.7

Experiment. The step-size sweep is reported in Section 5.4 (Table 5) and is not repeated here: all steps up to 5.95×10^{-4} are admissible, 6.0×10^{-4} reaches T with negative concentrations, and 7×10^{-4} , 8×10^{-4} diverge; a bisection places the boundary at $h_{\text{emp}} \approx 5.962 \times 10^{-4}$, i.e. 1.1% above h_{crit} . Table 2 sets the local prediction beside these outcomes. In both diverging runs the blow-up occurs about nine time units after the local limit is first exceeded (8.9 and 9.1), which is plausible because a mildly unstable mode needs time to grow from round-off and truncation errors. For $h = 6 \times 10^{-4}$ the limit is exceeded only from $t \approx 37.0$, i.e. too late for a comparable delay to end in divergence before $T = 40$, in agreement with the observed finite but negative solution. The same reasoning explains why $h = 5.9 \times 10^{-4}$ and 5.95×10^{-4} , slightly above h_{crit} , are still admissible. This delay was measured on two runs only, so it supports the picture but does not prove it.

Interpretation and limitations. The agreement should be read as empirical support for a local estimate, not as a nonlinear stability theorem: the Jacobian varies along the trajectory, and the scalar argument describes only the linearised amplification. Non-normality is mild where the constraint is decided ($\kappa \lesssim 2.2$ at late times). Finally, linear stability does not by itself guarantee positivity or conservation: at $h = 6 \times 10^{-4}$ the invariant defect is still at round-off level while the solution is already negative, so the three properties have to be tested separately (Section 5.4).

5 Numerical Experiments and Validation

This section asks whether the numerical results for the Robertson problem can be trusted, and what they reveal about stiffness. Three questions structure the experiments: (i) are the implementations correct, i.e. do they reproduce their formal orders against a verified reference? (ii) which step-size restriction is active for each method, accuracy or stability, and is the solution still physically admissible? (iii) what does each method cost at comparable accuracy? Stability, conservation, non-negativity and accuracy are treated as *separate* properties: a trajectory can be finite and conserve $y_1 + y_2 + y_3 = 1$ while being inaccurate or negative. The numerical comparison follows the standard distinction between methods intended for nonstiff and stiff ODE systems [11, 12]. In particular, stiff-oriented implicit algorithms are designed to avoid the severe stability restrictions that can dominate explicit integration.

The four fixed-step integrators, adaptive Explicit Euler, and adaptive Implicit Euler with first-order and locally extrapolated variants are implemented from scratch using NumPy arrays [13], with the time-stepping loops compiled using Numba [14]; Newton’s method uses the analytic Jacobian and a dense 3×3 linear solve. All numerical figures were generated using Matplotlib [15]. SciPy’s Radau and BDF [16, 17] are used only as the independent oracle for the reference and, in Section 5.6, as a library baseline. All errors use

$$E_\infty = \max_n \|\mathbf{y}_n - \mathbf{y}_{\text{ref}}(t_n)\|_2, \quad (49)$$

where t_n runs over *every* time point stored by the method and $\mathbf{y}_{\text{ref}}$ is evaluated by dense output. The accuracy of that dense output is checked in Section 5.1.

5.1 Reference Solution Construction

The reference is SciPy’s Radau method [16, 5] with $\text{rtol} = 10^{-12}$, $\text{atol} = 10^{-14}$. Two independent checks support it: the run is repeated with tolerances 100 times looser, and a BDF run with the same tight tolerances is compared with it. Both comparisons are made at the 201 prescribed output times ($t = 0$ and 200 log-spaced points in $[10^{-8}, 40]$) and on a fine grid of 60,000 points, which tests the dense output used in Eq. (49).

Table 3: Validation checks for the reference solution (maximum Euclidean discrepancy).

Check	Compared at	Discrepancy
Radau, 100 * looser	201 output times	6.30×10^{-12}
Radau vs. BDF (same tolerances)	201 output times	1.94×10^{-11}
Radau, 100 * looser	60,000 fine-grid points (dense output)	1.03×10^{-11}
Radau vs. BDF	60,000 fine-grid points (dense output)	1.95×10^{-11}
Invariant defect $\max y_1 + y_2 + y_3 - 1 $	201 output times	1.33×10^{-15}

All discrepancies are about 2×10^{-11} or smaller (Table 3), so only nine decimals are retained:

$$\mathbf{y}_{\text{ref}}(40) \approx (0.715827069, 9.1855 \times 10^{-6}, 0.284163746).$$

The finest RK4 error is approximately 1.2×10^{-10} , while the finest extrapolated IE error is 9.1×10^{-11} . These are only a few times the Radau–BDF discrepancy. They indicate proximity to the reference, but their final digits and any apparent ordering at that scale are not independently certified.

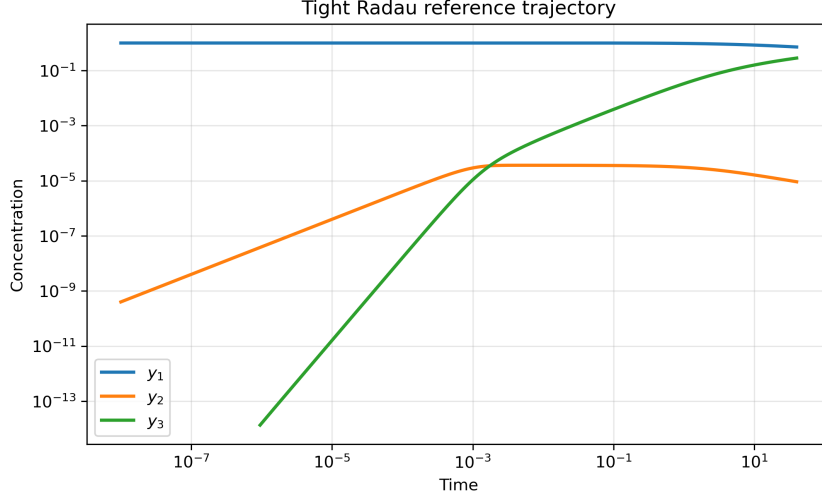


Figure 7: Tight Radau reference trajectory (values below $\text{atol} = 10^{-14}$ are not drawn). y_2 stays several orders of magnitude below y_1 for all t , and below y_3 for $t \gtrsim 2 \times 10^{-3}$.

5.2 Sanity Checks

Before interpreting convergence, stability, and efficiency results, we verify several problem-specific sanity checks. These checks reuse the reference-validation, physical-admissibility, and Newton-tolerance experiments reported in this section.

The reference solution is independently supported by Radau tolerance refinement, a Radau–BDF comparison, and the invariant defect reported in Table 3. In particular,

$$\max_t |y_1(t) + y_2(t) + y_3(t) - 1| = 1.33 \times 10^{-15},$$

and

$$y_{\text{ref}}(40) \approx (0.715827069, 9.1855 \times 10^{-6}, 0.284163746).$$

Physical admissibility is checked separately from accuracy. The fixed-step runs in Table 4 remain non-negative, whereas the Explicit Euler boundary study in Table 5 shows that a step size of 6.0×10^{-4} already produces negative concentrations even though the invariant defect remains near round-off. This confirms that conservation alone is not sufficient to establish a physically valid solution. The expected transient behaviour is also observed: y_2 remains small, peaks early, and subsequently decays to a small positive value at $t = 40$.

For the implicit methods, Table 6 verifies that the Newton tolerance is sufficiently strict. Loose tolerances produce non-physical solutions or Newton failure, while sufficiently tight tolerances prevent nonlinear-solver error from dominating the reported time-discretisation error. The invariant defect is not informative here: it stays at round-off for every tolerance (Section 5.5.1).

5.3 Convergence Order Verification

Validation protocol. Because no exact trajectory is available, correctness rests on three independent checks. (i) *Independent oracle*: the Radau reference, verified by a tolerance refinement and by BDF (Section 5.1). (ii) *Known properties*: the linear invariant $y_1 + y_2 + y_3 = 1$ and non-negativity, tracked separately from accuracy (Section 5.4). (iii) *Cross-order self-consistency*:

methods of orders 1, 2 and 4 must converge to the same limit with their formal orders. At their finest step sizes the RK4 and Trapezoidal/CN runs differ from the reference by 1.2×10^{-10} and 4.9×10^{-8} , so by the triangle inequality they agree with each other to within 5×10^{-8} ; the two Euler methods, of first order, reach 6.7×10^{-7} (explicit) and 1.3×10^{-5} (implicit) at their finest steps, consistent with their orders. The slopes are extracted below.

For a method of formal order p , $E(h) \approx Ch^p$, so the observed pairwise order between successive step sizes is

$$p_{\text{obs}} = \frac{\log(E_i/E_{i+1})}{\log(h_i/h_{i+1})}. \quad (50)$$

Each method is run with four step sizes, each half of the previous one (three successive halvings), inside its stable range. Explicit Euler starts at $h = 5 \times 10^{-4}$, just below its stability limit (Section 5.4); RK4 starts at $h = 8 \times 10^{-4}$, which is already 97% of its own real-axis limit $h \lesssim 8.21 \times 10^{-4}$ (the RK4 stability interval is $[-2.785, 0]$). The implicit runs solve the nonlinear system to $\tau_N = 10^{-13}$ (Section 5.5.1).

Table 4: Four-method convergence study. All 16 runs reached $T = 40$ with finite values (stable in this sense); “Non-neg.” states whether $\min_{n,j} y_{j,n} \geq -10^{-12}$; p_{fit} is the least-squares slope over the three finest step sizes.

Method	h	E_∞	Pairwise order	Non-neg.	p_{fit}
Explicit Euler	5.00×10^{-4}	6.831×10^{-6}	–	yes	1.060
	2.50×10^{-4}	2.893×10^{-6}	1.240	yes	
	1.25×10^{-4}	1.365×10^{-6}	1.084	yes	
	6.25×10^{-5}	6.655×10^{-7}	1.036	yes	
RK4	8.00×10^{-4}	1.172×10^{-6}	–	yes	4.247
	4.00×10^{-4}	4.377×10^{-8}	4.743	yes	
	2.00×10^{-4}	2.179×10^{-9}	4.329	yes	
	1.00×10^{-4}	1.215×10^{-10}	4.165	yes	
Implicit Euler	1.00×10^{-2}	1.039×10^{-4}	–	yes	1.000
	5.00×10^{-3}	5.200×10^{-5}	0.999	yes	
	2.50×10^{-3}	2.601×10^{-5}	1.000	yes	
	1.25×10^{-3}	1.301×10^{-5}	1.000	yes	
Trapezoidal / CN	1.00×10^{-3}	1.470×10^{-6}	–	yes	2.004
	5.00×10^{-4}	7.825×10^{-7}	0.910	yes	
	2.50×10^{-4}	1.949×10^{-7}	2.005	yes	
	1.25×10^{-4}	4.864×10^{-8}	2.003	yes	

Using one rule for all methods—the least-squares slope over the three finest step sizes—gives $p_{\text{EE}} \approx 1.06$, $p_{\text{RK4}} \approx 4.25$, $p_{\text{IE}} \approx 1.000$ and $p_{\text{CN}} \approx 2.004$ (Table 4, Figure 8). The coarsest pairs are pre-asymptotic: CN gives 0.910, and RK4’s first pair (4.74) lies close to its stability limit; the pairwise orders then decrease towards the formal values (Explicit Euler $1.24 \rightarrow 1.04$, RK4 $4.74 \rightarrow 4.17$). The table also records that every run is finite, reaches $T = 40$ and is non-negative, so the differences seen here are purely differences of accuracy.

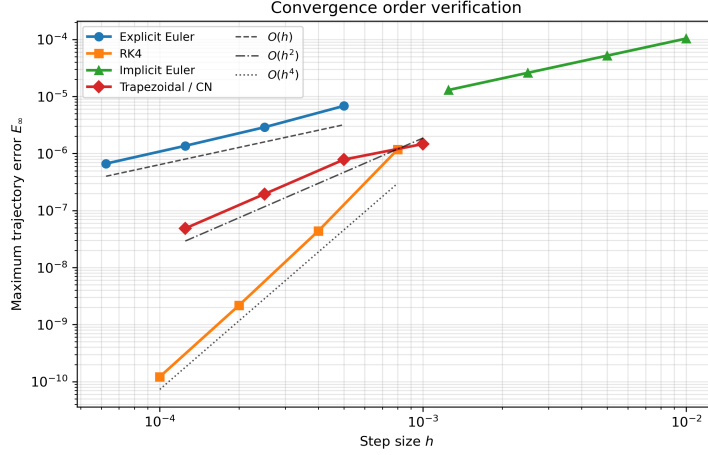


Figure 8: Convergence orders with respect to E_∞ . The dashed, dash-dotted and dotted lines are slope guides of order 1, 2 and 4.

5.4 Explicit Euler: Stability Boundary and Physical Admissibility

The eigenvalue analysis of Section 4.2 gives $|\lambda|_{\max} = 3.393 \times 10^3$, attained at the final time $t = 40$, and therefore the frozen-Jacobian estimate of Eq. (48)

$$h_{\text{crit}} = \frac{2}{|\lambda|_{\max}} \approx 5.895 \times 10^{-4}.$$

This is a local estimate, not a nonlinear stability theorem, so it is tested by a step-size sweep in which three diagnostics are kept apart: non-negativity, full-state error, and the invariant defect $\max_n |y_{1,n} + y_{2,n} + y_{3,n} - 1|$.

Table 5: Explicit Euler near the predicted limit h_{crit} . Admissible: reaches T with $\min y_j \geq -10^{-12}$; negative: reaches T but violates non-negativity; diverged: blow-up before T (time at which $\max_j |y_j|$ first exceeds 2). The invariant defect is evaluated only over steps with $\max_j |y_j| \leq 2$, because beyond that it measures round-off of huge numbers.

h	Status	Minimum y_j	Invariant defect	Final-time error
4.00×10^{-4}	admissible	0	1.07×10^{-14}	1.98×10^{-6}
5.00×10^{-4}	admissible	0	1.24×10^{-14}	2.47×10^{-6}
5.50×10^{-4}	admissible	0	2.62×10^{-14}	2.72×10^{-6}
5.90×10^{-4}	admissible	0	1.07×10^{-14}	2.91×10^{-6}
5.95×10^{-4}	admissible	0	1.20×10^{-14}	2.94×10^{-6}
6.00×10^{-4}	negative values	-4.84×10^{-6}	6.66×10^{-15}	4.48×10^{-3}
7.00×10^{-4}	diverged ($t \approx 27.1$)	—	6.66×10^{-15}	—
8.00×10^{-4}	diverged ($t \approx 17.7$)	—	4.33×10^{-15}	—

Table 5 and Figure 9 show that all steps up to $h = 5.95 \times 10^{-4}$ are admissible. At $h = 6.0 \times 10^{-4}$ the run still reaches $T = 40$, but concentrations become negative ($\min y_j = -4.8 \times 10^{-6}$) and the final error grows by three orders of magnitude to 4.5×10^{-3} ; at $h = 7 \times 10^{-4}$ and 8×10^{-4} it diverges at $t \approx 27$ and $t \approx 18$. A bisection between 5.9×10^{-4} (admissible) and 6.0×10^{-4} (non-admissible) places the admissibility boundary at $h_{\text{emp}} \approx 5.962 \times 10^{-4}$, i.e. 1.1% above h_{crit} (the bisection assumes a monotone transition inside this interval). The small excess is consistent with the local nature of the estimate: since $|\lambda_{\max}|$ is largest at the end, the local limit $2/|\lambda_{\max}(t)|$ is exceeded by h_{emp} only for $t \gtrsim 38.1$, so the weakly amplified fast mode has little time to grow (Table 2 in Section 4.3 lists these times for all step sizes). For this

trajectory the frozen-Jacobian value is thus an accurate and slightly conservative predictor; this is evidence, not a proof.

The three diagnostics fail independently. The invariant defect stays at round-off level [18] ($\sim 10^{-14}$) even at $h = 6.0 \times 10^{-4}$, where the solution is already negative and inaccurate: every explicit Runge–Kutta step preserves $\mathbf{1}^T \mathbf{y}$ because $\mathbf{1}^T \mathbf{f} \equiv 0$, so conservation carries no information about admissibility or accuracy. (For the diverged runs the defect is evaluated only while $\max_j |y_j| \leq 2$; beyond that it merely measures round-off of huge numbers.)

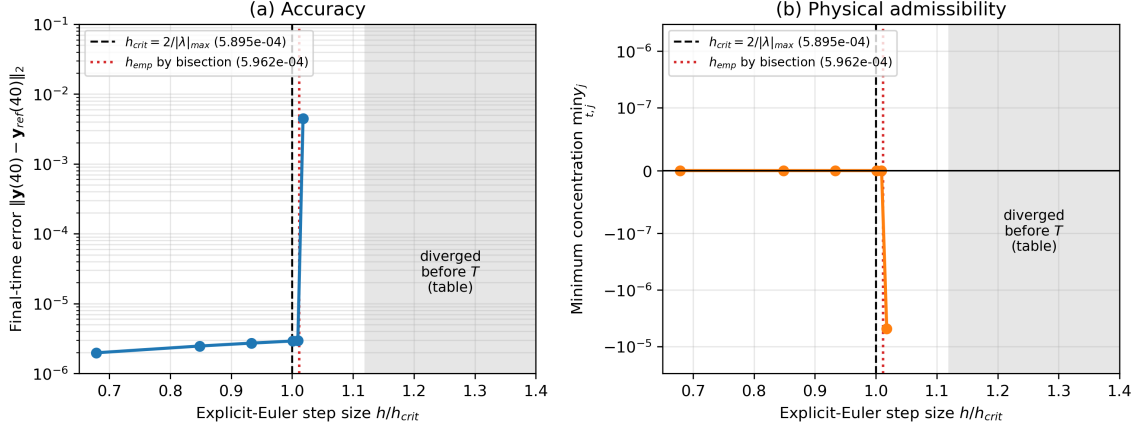


Figure 9: Explicit Euler near h_{crit} . (a) final-time error; (b) minimum concentration over the run. The step size is scaled by h_{crit} ; the shaded region contains the diverged runs $h = 7 \times 10^{-4}$, 8×10^{-4} (Table 5).

5.5 Solver Diagnostics

5.5.1 Newton-Tolerance Sensitivity and Initial Guess

Implicit Euler solves a nonlinear system in every step, so its error contains time-discretisation error and algebraic error. To separate them, $h = 10^{-2}$ is fixed and the stopping tolerance τ_N of the *absolute* residual test $\|F\|_\infty \leq \tau_N$ is varied. Because $y_2 = O(10^{-5})$ (peak 3.6×10^{-5}), τ_N has to be judged against y_2 , not against the $O(1)$ components.

Table 6: Newton-tolerance sensitivity of Implicit Euler at fixed $h = 10^{-2}$ (4000 steps). The stopping test is $\|F\|_\infty \leq \tau_N$ on the absolute residual; “Newton updates” counts linear solves summed over all steps.

τ_N	Status	$y_2(40)$	$ \Delta y_2 $	Final L^2 error	Newton updates	Max residual	Invariant defect
1×10^{-2}	non-physical	-3.591×10^{-6}	1.28×10^{-5}	3.63×10^1	7860	1.0×10^{-2}	1.3×10^{-13}
1×10^{-4}	failed ($t = 3.71$)	—	—	—	—	—	8.9×10^{-16}
1×10^{-6}	completed	9.191×10^{-6}	5.79×10^{-9}	5.44×10^{-5}	2095	1.0×10^{-6}	1.8×10^{-15}
1×10^{-8}	completed	9.187×10^{-6}	1.16×10^{-9}	4.41×10^{-5}	3427	1.0×10^{-8}	2.9×10^{-15}
1×10^{-10}	completed	9.187×10^{-6}	1.36×10^{-9}	4.94×10^{-5}	4012	9.4×10^{-12}	1.6×10^{-15}
1×10^{-12}	completed	9.187×10^{-6}	1.36×10^{-9}	4.94×10^{-5}	4014	1.1×10^{-13}	2.4×10^{-15}

Table 6 shows three regimes. For $\tau_N = 10^{-2}$ and 10^{-4} , which exceed the peak of y_2 , the solve cannot resolve y_2 : in both runs a component turns negative from $t \approx 0.02$ on ($\min y_j = -28.8$ and -8.0), and the $\tau_N = 10^{-4}$ run finally fails at $t \approx 3.71$, when Newton no longer converges within 50 iterations from the unphysical state. A loose tolerance therefore does not merely lose accuracy; it destroys admissibility, and the failure appears long after the damage was done. For $\tau_N \leq 10^{-6}$ the solution is physical, but the error is converged only for $\tau_N \leq 10^{-10}$: the final error is 5.44×10^{-5} (10^{-6}) and 4.41×10^{-5} (10^{-8}) against the converged 4.94×10^{-5} , and $|\Delta y_2|$ settles at 1.36×10^{-9} . These deviations are consistent with per-step residuals accumulating

over the 4000 steps (worst case $N\tau_N = 4 \times 10^{-5}$ for $\tau_N = 10^{-8}$), which suggests the rule $\tau_N \lesssim E_{\text{target}}/N_{\text{steps}}$. For the fixed-step study, with up to 6.4×10^5 steps, $\tau_N = 10^{-13}$ was chosen so that algebraic error cannot contaminate the order estimates; we did not test whether a looser value would suffice there.

The invariant defect does *not* diagnose an unconverged Newton solve: it is $\lesssim 10^{-13}$ for every τ_N in Table 6. Since $\mathbf{1}^T J \equiv 0$, a single Newton update with the exact Jacobian restores $\sum_j y_j$ to its value at the previous step, so residual size and defect are decoupled here; the residual, y_2 and the error must be monitored instead.

The initial guess matters as well. At $h = 10^{-2}$ and $\tau_N = 10^{-10}$, the explicit-Euler predictor and the previous solution both converge to the same physical solution ($E_\infty = 1.04 \times 10^{-4}$), whereas a deliberately perturbed predictor ($\Delta y_1 = +0.1$, $\Delta y_2 = -0.1$) passes the same residual test but converges to a non-physical root of the implicit equation ($E_\infty = 7.06$, 7652 negative entries). A small residual therefore certifies neither correctness nor admissibility.

5.5.2 Adaptive Step-Size Control

Both adaptive methods are validated by tolerance refinement against the same dense-output reference. Adaptive EE retains the two-half-step value. Adaptive IE is run both with this first-order value and with the local extrapolation in Eq. (47). Their common absolute error proxy and controller are given in Section 3.5; the different step ceilings are stated there explicitly.

Table 7: Adaptive Explicit Euler (step doubling, $h_0 = 10^{-4}$, $h_{\text{max}} = 10^{-2}$). The tolerance bounds the *local* error estimate per step; E_∞ is the accumulated *global* error.

Tol.	Status	Accepted	Rejected	E_∞	E_∞/tol	Mean accepted h
1×10^{-4}	diverged ($t \approx 0.06$)	416	40	—	—	—
1×10^{-5}	completed	22248	9794	1.02×10^{-3}	102	1.80×10^{-3}
3×10^{-6}	completed	25279	12544	1.32×10^{-4}	44	1.58×10^{-3}
1×10^{-6}	completed	25398	12686	1.97×10^{-5}	20	1.57×10^{-3}
3×10^{-7}	completed	25407	12692	1.20×10^{-5}	40	1.57×10^{-3}
1×10^{-7}	completed	23202	9194	1.35×10^{-5}	135	1.72×10^{-3}
3×10^{-8}	completed	25221	12145	1.24×10^{-5}	413	1.59×10^{-3}
1×10^{-8}	completed	23794	8854	9.67×10^{-6}	967	1.68×10^{-3}

Table 8: Adaptive step-doubling controllers on the same tolerances: Explicit Euler (Table 7), implicit Euler keeping the two-half-step value (first order) and implicit Euler with local extrapolation $2y_{\text{two}} - y_{\text{full}}$ (second order). “Acc.” counts accepted steps; all runs use $h_0 = 10^{-4}$ and the same absolute max-norm test $\|y_{\text{two}} - y_{\text{full}}\|_{\infty} \leq \text{tol}$; the implicit runs use $h_{\text{max}} = 5$ and a Newton tolerance of 10^{-13} .

Tol.	Explicit Euler		Implicit Euler, 1st order		Implicit Euler, extrapolated		
	Acc.	E_{∞}	Acc.	E_{∞}	Acc.	E_{∞}	E_{∞}/tol
1×10^{-4}	–	diverged	60	1.9×10^{-3}	60	8.0×10^{-5}	0.80
1×10^{-5}	22248	1.0×10^{-3}	165	6.1×10^{-4}	164	8.7×10^{-6}	0.87
3×10^{-6}	25279	1.3×10^{-4}	293	3.4×10^{-4}	292	2.7×10^{-6}	0.89
1×10^{-6}	25398	2.0×10^{-5}	502	2.0×10^{-4}	501	9.0×10^{-7}	0.90
3×10^{-7}	25407	1.2×10^{-5}	912	1.1×10^{-4}	911	2.7×10^{-7}	0.91
1×10^{-7}	23202	1.3×10^{-5}	1575	6.2×10^{-5}	1574	9.1×10^{-8}	0.91
3×10^{-8}	25221	1.2×10^{-5}	2873	3.4×10^{-5}	2872	2.7×10^{-8}	0.91
1×10^{-8}	23794	9.7×10^{-6}	4975	2.0×10^{-5}	4973	9.1×10^{-9}	0.91
1×10^{-9}	–	not run	15723	6.2×10^{-6}	15721	9.1×10^{-10}	0.91
1×10^{-10}	–	not run	49714	2.0×10^{-6}	49712	9.1×10^{-11}	0.91

Separate local/global scaling from an error plateau. The statement $E_{\infty}/\text{tol} > 1$ does not by itself imply failure or stiffness-induced saturation: the tolerance controls a local proxy, whereas E_{∞} is a global trajectory error. The first-order implicit variant illustrates this distinction: its global error decreases as the tolerance is tightened even though E_{∞}/tol increases (Table 8). Formally, $e_n = O(h^2)$ gives $h = O(\sqrt{\text{tol}})$ and a first-order global error $O(\sqrt{\text{tol}})$, so the ratio can grow without any error plateau.

For adaptive EE, a different observation matters: over $10^{-6} \geq \text{tol} \geq 10^{-8}$ its global error remains roughly 9.7×10^{-6} – 2.0×10^{-5} and its accepted-step count remains close to 2.5×10^4 . The steps stay near the explicit stability boundary, and about one third of trials are rejected. This is an empirical plateau over the tested range, not proof of a nonzero error floor for arbitrarily small tolerances. At $\text{tol} = 10^{-4}$ the explicit run diverges at $t \approx 0.06$; the absolute test is too loose to resolve the small intermediate species.

Our adaptive implicit results. The locally extrapolated IE variant completes all ten tested tolerances from 10^{-4} to 10^{-10} with non-negative concentrations. Its global ratio lies between 0.80 and 0.91, and its maximum invariant defect is 2.6×10^{-14} . At $\text{tol} = 1 \times 10^{-6}$ it achieves $E_{\infty} = 9.00 \times 10^{-7}$ with 501 accepted steps; at the finest tolerance 1×10^{-10} the measured error is 9.1×10^{-11} . The latter is only a few times the approximately 2×10^{-11} reference discrepancy and should not be interpreted as independently certified to all displayed digits.

The observed $E_{\infty} = O(\text{tol})$ behaviour is consistent with second-order global error under a controller that chooses $h = O(\sqrt{\text{tol}})$. It is an empirical scaling result for this problem, not a general promise that the global error is bounded by the local tolerance. The first-order implicit comparison isolates the benefit of extrapolation from the removal of the explicit stability restriction.

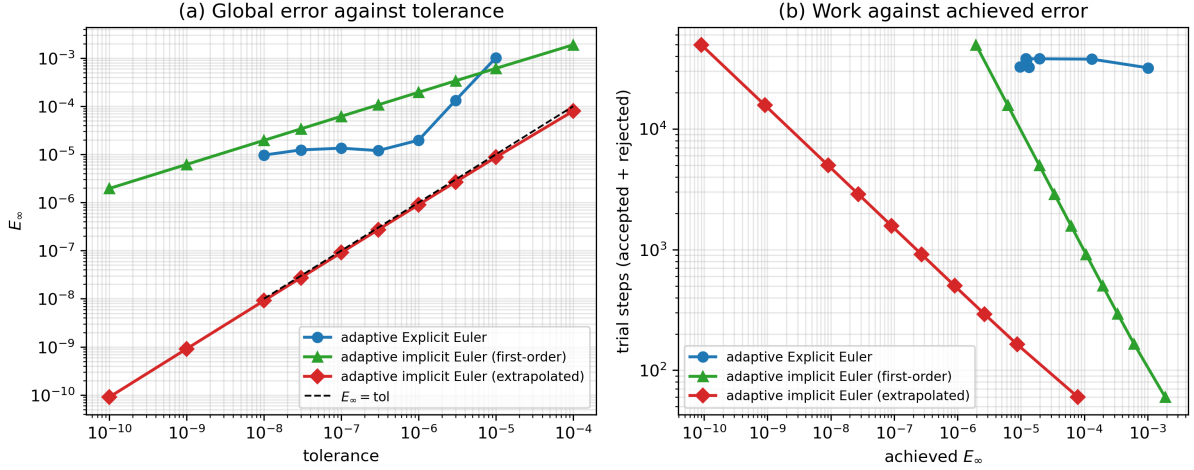


Figure 10: Tolerance refinement of adaptive EE and both adaptive IE variants. (a) Global error versus local tolerance. (b) Trial-step count versus achieved error. A trial step is not equal work across methods: IE uses three nonlinear solves, whereas EE uses two right-hand-side evaluations.

5.6 Efficiency Comparison

Efficiency is compared using achieved global error, rather than nominal tolerance. Runtime is the median of seven warmed calls, excluding compilation and plotting. The fixed-step study is retained, and the same timing protocol is applied to adaptive EE and both adaptive IE variants. Counts include rejected trials: EE uses two right-hand-side evaluations per trial, while IE performs three nonlinear substep solves. Newton-update, right-hand-side and step counts describe different operations and should not be equated.

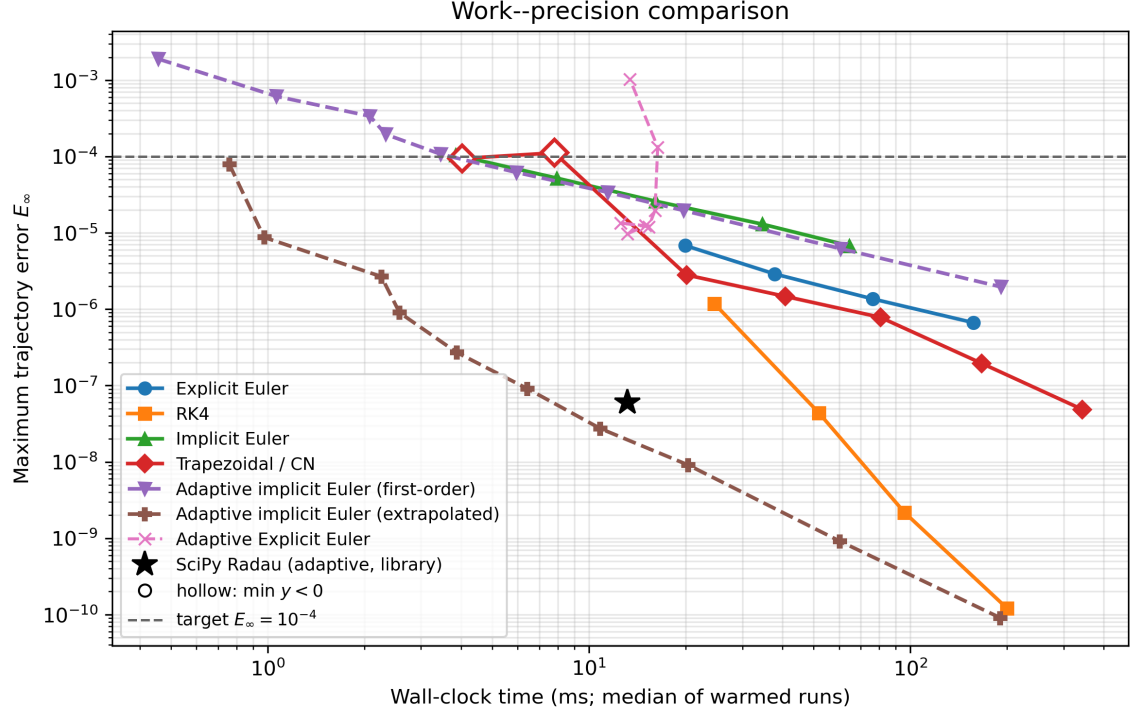


Figure 11: Work-precision diagram including the four fixed-step methods, adaptive EE, and both self-implemented adaptive IE variants. Hollow markers flag negative concentrations in the fixed-step runs. The star is SciPy Radau with $\text{rtol} = 10^{-6}$, $\text{atol} = 10^{-9}$, retained as a library baseline.

Table 9: Cost at comparable accuracy (median of warmed runs, this machine). Times are implementation- and machine-dependent; the SciPy row is not JIT-compiled and its time is not like-for-like with the compiled kernels, so compare its work counts.

Method	h	E_∞	Min. y_j	Time (ms)	Work
<i>(a) Target $E_\infty \leq 10^{-4}$: fastest tested run per method</i>					
Explicit Euler	5.00×10^{-4}	6.83×10^{-6}	≥ 0	19.9	80000 steps; 80000 RHS evaluations
Implicit Euler	5.00×10^{-3}	5.20×10^{-5}	≥ 0	8.0	8000 steps; 8014 Newton updates
RK4	8.00×10^{-4}	1.17×10^{-6}	≥ 0	24.6	50000 steps; 200000 RHS evaluations
Trapezoidal / CN	2.00×10^{-3}	2.81×10^{-6}	≥ 0	20.1	20000 steps; 20015 Newton updates
Trapezoidal / CN (non-physical)	1.00×10^{-2}	9.41×10^{-5}	-2.9×10^{-5}	4.0	4000 steps; 4102 Newton updates
<i>(b) Matched accuracy $E_\infty \approx 7 \times 10^{-6}$</i>					
Explicit Euler	5.00×10^{-4}	6.83×10^{-6}	≥ 0	19.9	80000 steps; 80000 RHS evaluations
Implicit Euler	6.50×10^{-4}	6.76×10^{-6}	≥ 0	64.6	61539 steps; 61558 Newton updates
<i>(c) Self-implemented adaptive Euler methods</i>					
Adaptive IE, extrapolated, tol= 1×10^{-4}	–	7.95×10^{-5}	≥ 0	0.8	60 steps (0 rejected); 370 Newton updates
Adaptive IE, first-order, tol= 1×10^{-7}	–	6.19×10^{-5}	≥ 0	5.9	1575 steps (2 rejected); 4800 Newton updates
Adaptive IE, extrapolated, tol= 1×10^{-5} (near-matched error)	–	8.73×10^{-6}	≥ 0	1.0	164 steps (0 rejected); 994 Newton updates
Adaptive EE, tol= 1×10^{-7}	–	1.35×10^{-5}	≥ 0	12.5	23202 steps (9194 rejected); 64792 RHS evaluations
<i>(d) Library reference: adaptive implicit solver</i>					
SciPy Radau, rtol= 10^{-6}	–	6.05×10^{-8}	≥ 0	13.1	66 steps; 545 RHS evaluations; 14 Jacobians

Fixed-step comparison. At the target $E_\infty \leq 10^{-4}$, Explicit Euler is forced to use a stability-limited step and its fastest tested admissible point is 15 times more accurate than required. Fixed-step Implicit Euler meets the target with $h = 5 \times 10^{-3}$. RK4 improves accuracy within its stability interval, whose real-axis limit 8.21×10^{-4} is only about 1.4 times the Explicit-Euler limit. CN at $h = 10^{-2}$ meets the error target but has negative concentrations; physical admissibility therefore remains a separate constraint.

At matched fixed-step accuracy $E_\infty \approx 7 \times 10^{-6}$, Implicit Euler requires 61539 steps and takes 3.2 times the Explicit-Euler time on this machine. At about $1.2\text{--}1.4 \times 10^{-6}$ error, RK4 uses 50 000 steps and 200000 right-hand-side evaluations, compared with 320000 Explicit-Euler steps; the measured speed ratio is 3.1. These statements concern the fixed-step implementations and do not establish a ranking against adaptive IE.

Self-implemented adaptive IE, adaptive EE and Radau. Among the tested extrapolated IE runs meeting $E_\infty \leq 10^{-4}$ and non-negativity, the fastest uses tol = 1×10^{-4} , achieves $E_\infty = 8.0 \times 10^{-5}$ in 60 accepted steps and 370 Newton updates, and takes 0.8 ms. It is 21.7

times faster than the fastest qualifying fixed-step IE run on this machine. At approximately the fixed-step EE matched accuracy, extrapolated IE with $\text{tol} = 1 \times 10^{-5}$ reaches $E_\infty = 8.7 \times 10^{-6}$ using 164 steps and 994 updates. The two measured errors are close, not identical; this row should be read as a near-matched comparison.

At the shared nominal absolute tolerance 10^{-6} , adaptive EE requires 25 398 accepted steps with global error 1.97×10^{-5} , while extrapolated IE requires 501 with error 9.00×10^{-7} . The first-order IE curve shows that implicit stepping alone removes the severe explicit step restriction but does not provide the extrapolated variant’s accuracy. Figure 11 therefore compares the full measured error–cost curves rather than treating equal tolerance as equal accuracy.

SciPy Radau reaches $E_\infty \approx 6.0 \times 10^{-8}$ in 66 accepted steps and 545 right-hand-side evaluations. It remains an independent library baseline: its fifth-order formula, relative-plus-absolute error scaling and implementation differ from our absolute-norm step-doubling methods. Its steps contain different work, and its library wall time is not a controlled comparison with our Numba kernels. The results support an efficiency advantage for our extrapolated IE at moderate accuracy, while higher-order fixed-step methods remain competitive at tighter achieved errors; no universal timing crossover is claimed.

Summary of findings. The reference and original fixed-step convergence results are preserved. The adaptive study now distinguishes three effects: explicit stability-limited steps, first-order implicit global-error accumulation, and the accuracy gain from local extrapolation. Conservation, non-negativity, nonlinear residual and global error remain separate diagnostics. Cost comparisons must specify achieved error, physical admissibility and implementation, rather than rely on nominal tolerance or raw step count alone.

6 Discussion

6.1 Physical Interpretation via Quasi-Steady-State Approximation (QSSA)

The numerical results of Sections 4 and 5 raise two questions that a purely numerical study cannot answer: why does the intermediate species B stay so small, and why does the problem behave as a stiff one even though B is almost absent? The reference solution (Figure 7) shows y_2 climbing from zero to about 3.6×10^{-5} within a few 10^{-3} time units, remaining near this level for more than two decades in time and then decaying slowly to $y_2(40) = 9.1855 \times 10^{-6}$, although B is consumed with rate constants up to $k_2 = 3 \times 10^7$. This section explains the magnitude and the time structure of y_2 by a quasi-steady-state argument [19, 20], makes it rigorous with a singular-perturbation (Tikhonov) framework, and connects it to the eigenvalue structure of Section 4.2 and to the numerical findings of Section 5.

Order-of-magnitude balance. The equation for the intermediate species is $\dot{y}_2 = k_1 y_1 - k_3 y_2 y_3 - k_2 y_2^2$. At early times $y_1 \approx 1$ and $y_3 \approx 0$, so $\dot{y}_2 \approx k_1 - k_2 y_2^2$: B is produced slowly (rate $k_1 = 0.04$) and consumed by a quadratic term with an enormous coefficient. Setting $\dot{y}_2 = 0$ gives the balance $k_1 y_1 \approx k_2 y_2^2$, i.e.

$$y_2^* \approx \sqrt{\frac{k_1 y_1}{k_2}} = \sqrt{\frac{0.04}{3 \times 10^7}} \approx 3.65 \times 10^{-5}. \quad (51)$$

The reference has $\max_t y_2 = 3.6487 \times 10^{-5}$, attained at $t = 4.56 \times 10^{-3}$, only 0.08% below Eq. (51). B stays small because the quadratic sink reaches the production rate at a tiny concentration, and $y_2^* \propto k_2^{-1/2}$ decreases as the sink becomes stronger. Two points refine the picture “a spike followed by a rapid relaxation”. First, there is no overshoot: $\max_t y_2$ lies slightly *below* y_2^* , so the rise is a monotone approach to the quasi-steady value. Second, what

follows the rise is not a fast relaxation but a slow decay on the time scale of the slow variables, along which y_2 follows its quasi-steady value.

One balance, two regimes. As y_3 grows, the sink $k_3 y_2 y_3$ takes over from $k_2 y_2^2$ and the balance becomes $k_1 y_1 \approx k_3 y_2 y_3$, i.e. $y_2^* \approx k_1 y_1 / (k_3 y_3)$. Both are limits of the single algebraic balance $k_2 y_2^2 + k_3 y_3 y_2 = k_1 y_1$, whose positive root is

$$y_2^{\text{QSSA}}(y_1, y_3) = \frac{-k_3 y_3 + \sqrt{k_3^2 y_3^2 + 4k_1 k_2 y_1}}{2k_2}. \quad (52)$$

Table 10 and Figure 12b compare the three predictions with the reference, evaluated with the reference values of y_1, y_3 . The early balance is accurate at $t = 10^{-2}$ (error at most 0.16% on $5 \times 10^{-3} < t < 10^{-2}$) but is 17% off at $t = 1$ and a factor 3.4 too large at $t = 40$; the late balance is the opposite and is 9.7% off at $t = 40$ (the value 1.01×10^{-5} against 9.1855×10^{-6}). Eq. (52) is accurate throughout: its relative error is below 9.2×10^{-5} for all $t \geq 5 \times 10^{-3}$ and 3.2×10^{-6} at $t = 40$.

Table 10: Quasi-steady predictions of y_2 divided by the reference value $y_2(t)$, evaluated with the reference y_1, y_3 . Early balance: $\sqrt{k_1 y_1 / k_2}$; late balance: $k_1 y_1 / (k_3 y_3)$; quadratic QSSA: positive root of $k_2 y_2^2 + k_3 y_3 y_2 = k_1 y_1$.

t	reference y_2	early balance	late balance	quadratic QSSA
1×10^{-2}	3.645×10^{-5}	1.0016	302	0.99991
1	3.075×10^{-5}	1.17	3.75	0.99993
40	9.186×10^{-6}	3.36	1.10	1.00000

Initial layer. For $t \lesssim 10^{-3}$ one has $y_1 \approx 1$ and $y_3 \approx 0$, and the equation $\dot{y}_2 = k_1 - k_2 y_2^2$ with $y_2(0) = 0$ can be solved exactly:

$$y_2(t) = \sqrt{\frac{k_1}{k_2}} \tanh(\sqrt{k_1 k_2} t), \quad t_f = \frac{1}{\sqrt{k_1 k_2}} = 9.13 \times 10^{-4}. \quad (53)$$

For small t this reduces to $y_2 \approx k_1 t$, the linear growth already seen in Section 4.2. Eq. (53) predicts that y_2 reaches 99% of y_2^* at $t = 2.42 \times 10^{-3}$; the reference gives 2.43×10^{-3} . Its relative error is at most 3.7×10^{-5} for $t \leq 10^{-3}$ (Figure 12a). The “spike” is thus a boundary layer of thickness $\sim t_f$, three to four orders of magnitude shorter than the slow time scale $1/k_1 = 25$. Adding the layer correction to Eq. (52), $y_2 \approx y_2^{\text{QSSA}} - y_2^*(0) [1 - \tanh(t/t_f)]$, brings the relative error below 9.2×10^{-5} on the whole interval (Figure 12c).

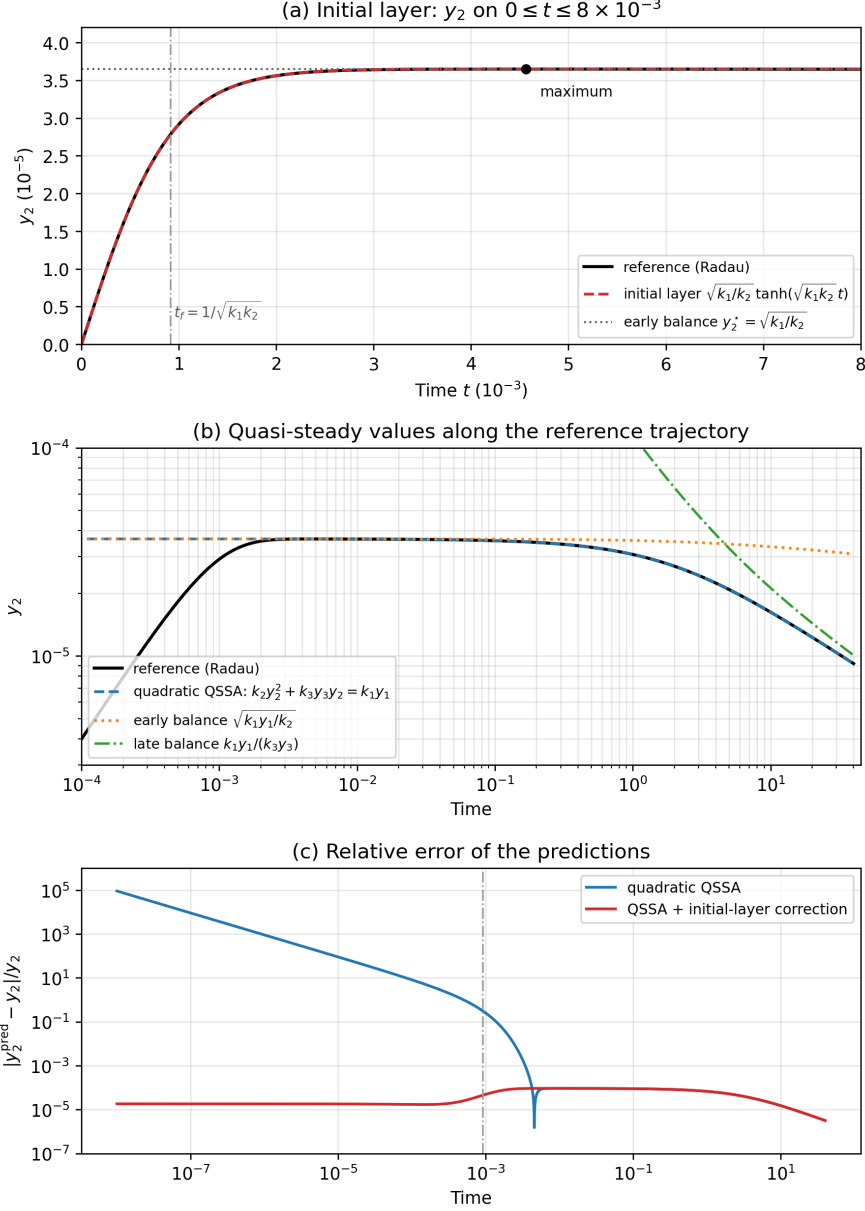


Figure 12: Quasi-steady-state analysis of y_2 . (a) Initial layer on $0 \leq t \leq 8 \times 10^{-3}$: reference, Eq. (53) and the early balance y_2^* . (b) Reference against the early balance, the late balance and the quadratic QSSA, Eq. (52). (c) Relative error of the quadratic QSSA and of the QSSA with initial-layer correction; the dash-dotted line marks t_f . All predictions use the reference y_1, y_3 .

Singular-perturbation framework (Tikhonov). A naive choice of the small parameter $\varepsilon = 1/k_2$ gives $\varepsilon \dot{y}_2 = \varepsilon(k_1 y_1 - k_3 y_2 y_3) - y_2^2$, whose critical manifold $y_2 = 0$ is degenerate ($\partial g / \partial y_2 = -2y_2 = 0$ there), so Tikhonov's theorem does not apply; this reflects that the quasi-steady value is $O(\sqrt{\varepsilon})$, not $O(\varepsilon)$. The remedy is to rescale $y_2 = \delta u$ with

$$\delta = k_2^{-1/2} = 1.83 \times 10^{-4}, \quad \kappa_3 = k_3 \delta = 1.83 = O(1),$$

where $k_3 \approx 10^4$ is regarded as being of order $k_2^{1/2}$. Since $k_2 \delta^2 = 1$, the system takes the standard slow-fast form

$$\dot{y}_1 = -k_1 y_1 + \kappa_3 u y_3, \quad \dot{y}_3 = u^2, \quad \delta \dot{u} = k_1 y_1 - \kappa_3 u y_3 - u^2 =: g(y_1, y_3, u), \quad (54)$$

with the invariant $y_1 + y_3 + \delta u = 1$. Tikhonov's theorem [21, 22] requires three conditions, all satisfied here:

- (i) *Critical manifold.* $g = 0$ has the isolated positive root $u = \varphi(y_1, y_3) = \frac{1}{2}[-\kappa_3 y_3 + \sqrt{\kappa_3^2 y_3^2 + 4k_1 y_1}]$, which is exactly Eq. (52) divided by δ .
- (ii) *Normal attraction.* $\partial g / \partial u = -2u - \kappa_3 y_3$, which on the manifold equals $-\sqrt{\kappa_3^2 y_3^2 + 4k_1 y_1} \leq -2\sqrt{k_1 y_1} < 0$ uniformly along the trajectory ($y_1 \geq 0.71$).
- (iii) *Initial value in the domain of attraction.* With $\tau = t/\delta$, the boundary-layer equation is $du/d\tau = k_1 - u^2$ (for $y_1 = 1, y_3 = 0$); every $u > -\sqrt{k_1}$ is attracted to $\sqrt{k_1}$, and $u(0) = 0$ qualifies. Its solution is Eq. (53).

The theorem then states that, for every fixed $t_1 > 0$, $(y_1, y_3) = (\bar{y}_1, \bar{y}_3) + O(\delta)$ and $u = \varphi(\bar{y}_1, \bar{y}_3) + O(\delta)$ uniformly on $[t_1, T]$, where $(\bar{y}_1, \bar{y}_3)$ solves the reduced system obtained by inserting $u = \varphi$ into the first two equations of (54). The deviation from the manifold decays like $\exp(-\mu t/\delta)$ with $\mu = |\partial g / \partial u|$, i.e. within a time $O(\delta)$ (t_f in Eq. (53)).

We checked the statement quantitatively. The reduced model (solved with Radau, without any use of the full solution) differs from the full solution by at most 1.3×10^{-5} in y_1 and 3.6×10^{-5} in y_3 for $t \geq 10^{-2}$; both are below $\delta = 1.83 \times 10^{-4}$, as predicted. The offset in y_3 is not random: the reduced model starts on the manifold, so it produces y_3 too early, by $k_2 \int_0^\infty [y_2^{*2} - y_2^2] dt = k_1 t_f = 3.65 \times 10^{-5}$, which is what is observed. This is the boundary-layer correction of the outer solution.

Link to the eigenvalue structure of Section 4.2. Condition (ii) also gives the fast eigenvalue: on the manifold $\lambda_{\text{fast}} = (\partial g / \partial u) / \delta = -(2k_2 y_2 + k_3 y_3)$, i.e.

$$\lambda_{\text{fast}} = -\sqrt{k_3^2 y_3^2 + 4k_1 k_2 y_1}. \quad (55)$$

This explains the three stages seen in Figure 4. (i) *Sharp early separation:* before y_2 reaches its quasi-steady value, $\lambda_{\text{fast}} \approx -2k_2 y_2 \approx -2k_1 k_2 t$ grows linearly in t , so the stiffness ratio rises from 1 to about 5×10^3 by $t \approx 10^{-3}$, i.e. in the initial layer. (ii) *Plateau:* once y_2 is slaved, $\lambda_{\text{fast}} \approx -2\sqrt{k_1 k_2} = -2.191 \times 10^3$ ($y_3 \approx 0$), against the computed 2.190×10^3 at $t = 10^{-2}$. The fast rate saturates near 2×10^3 , and not at $k_2 = 3 \times 10^7$, because k_2 multiplies the tiny y_2 , not 1; this is the reason for the remark that $|\lambda|_{\text{max}} = O(10^3)$. (iii) *Late milder growth:* as y_3 increases, Eq. (55) rises only to 3.3928×10^3 at $t = 40$ (computed 3.3928×10^3). Over more than four decades in time the fast eigenvalue therefore changes by a factor of about 1.5; the stiffness ratio nevertheless grows again to 1.6×10^5 because the slow eigenvalue decays from 0.4 to 0.02. Eq. (55) also gives the explicit-Euler limit in closed form: $h_{\text{crit}} = 2/\sqrt{k_3^2 y_3(T)^2 + 4k_1 k_2 y_1(T)} = 5.895 \times 10^{-4}$, identical to Eq. (48).

Consequences for the numerical results.

- *Stiffness is a stability, not an accuracy, restriction* (Sections 5.4 and 5.6). After about $3t_f \approx 3 \times 10^{-3}$, i.e. less than 0.01% of the interval, the fast mode carries no signal: y_2 is a slaved variable. Accuracy needs to resolve only the slow dynamics, but Explicit Euler must still keep $|1 + h\lambda_{\text{fast}}| < 1$ for the (numerically excited) fast mode, hence $h < 5.895 \times 10^{-4}$; this is why its best run is about 15 times more accurate than required. The adaptive controller of Section 3.5 cannot escape this limit either (Section 5.5.2).
- *L-stability decides what happens to departures from the manifold* (Section 4.1). For Implicit Euler $R(z) \rightarrow 0$: a deviation from the slow manifold is removed within one step, and the method effectively integrates the reduced model, which is why $h = 10^{-2}$ is admissible.

For Trapezoidal/CN $R(z) \rightarrow -1$: an unresolved deviation is not damped but reflected with alternating sign. At $h = 10^{-2}$ and 5×10^{-3} the layer, of duration t_f , is not resolved, and the observed negative concentrations ($\min y_j = -2.9 \times 10^{-5}$ and -4.2×10^{-5} , Section 5.6) have the size of $y_2^* = 3.65 \times 10^{-5}$ itself, which is what an undamped, unresolved layer perturbation of amplitude $\sim y_2^*$ would produce. We regard this as consistent with, but not a proof of, the mechanism.

- *Tolerances must resolve y_2^** (Sections 5.5.1 and 5.5.2). For an implicit step the Newton solve determines y_2 through the algebraic balance $g = 0$. An absolute residual tolerance of 10^{-2} or 10^{-4} , larger than y_2^* , cannot determine y_2 , which is why those runs turned negative; the adaptive run with tolerance 10^{-4} , also larger than the peak of y_2 , diverged.

Limits of the picture. The quasi-steady description is valid only after the layer: at $t = 2 \times 10^{-3}$ the QSSA alone is still 2.5% off. Tikhonov's theorem guarantees only $O(\delta)$ accuracy; the observed relative accuracy of about 10^{-4} after the correction is much better than this bound, which we do not claim to have explained. Finally, Figure 12 and Table 10 use the reference values of y_1, y_3 and are therefore diagnostics of the balance; the independent test of the reduced model is the comparison above.

6.2 Why Explicit Methods Fail and Implicit Methods Succeed

The evidence of Sections 4–5 forms one causal chain.

Cause. After the initial layer of duration $t_f = 9.13 \times 10^{-4}$ the fast mode is slaved to the slow dynamics (Section 6.1), but it stays in the Jacobian: $|\lambda_{\text{fast}}| = \sqrt{k_3^2 y_3^2 + 4k_1 k_2 y_1}$ rises from 2.2×10^3 to 3.393×10^3 , while the slow eigenvalue is only 0.02–0.4 and the stiffness ratio reaches 1.6×10^5 (Section 4.2).

Prediction. The stability regions of Explicit Euler and RK4 are bounded, with real-axis intervals $[-2, 0]$ and $[-2.7853, 0]$ (Section 4.1). With $|\lambda|_{\text{max}}$ attained at $t = 40$ this gives $h < 5.895 \times 10^{-4}$ and $h < 8.21 \times 10^{-4}$ (Section 4.3). Implicit Euler and Trapezoidal/CN are A-stable and impose no linear restriction; only Implicit Euler is also L-stable ($R \rightarrow 0$ instead of $R \rightarrow -1$).

Test. The sweep of Section 5.4 confirms the prediction: Explicit Euler is admissible up to $h_{\text{emp}} = 5.962 \times 10^{-4}$ (1.1% above h_{crit}), becomes negative at 6×10^{-4} and diverges at 7×10^{-4} and 8×10^{-4} , about nine time units after the local limit is first exceeded (Table 2). RK4 at $h = 8 \times 10^{-4}$ (97% of its limit) is stable and accurate, but its limit is only about 1.4 times that of Explicit Euler: its fourth order improves the accuracy inside the stable range, not the admissible step. The adaptive explicit controller rides the same limit, at the price of about one third rejected steps (Sections 3.5 and 5.5.2).

Consequence. Fixed-step explicit methods may be much more accurate than the target requires because stability dictates their step. Implicit Euler can take larger steps, but its first-order accuracy makes the fixed-step comparison depend on the error target. Our adaptive extrapolated IE combines implicit substeps with a second-order accepted value: at $\text{tol} = 1 \times 10^{-6}$ it achieves $E_\infty = 9.00 \times 10^{-7}$ in 501 accepted steps. The first-order adaptive IE control experiment separates this extrapolation gain from the removal of the explicit stability restriction. Section 5.6 compares their measured cost at achieved errors; no method is declared universally cheapest.

The Trapezoidal prediction. Section 4.1 predicted that Trapezoidal/CN, although A-stable, reflects very fast modes ($R \rightarrow -1$) instead of damping them. With $\lambda_{\text{fast}} \approx -2.19 \times 10^3$, the growth factor at $h = 10^{-3}, 2 \times 10^{-3}, 5 \times 10^{-3}, 10^{-2}$ is $R_{\text{CN}} \approx -0.05, -0.37, -0.69, -0.83$, whereas $R_{\text{IE}} \approx 0.31, 0.19, 0.08, 0.04$: a perturbation left by an unresolved layer is nearly preserved with alternating sign by CN and removed by Implicit Euler. This is consistent with what Section 5.6 observes: CN has negative concentrations at $h = 5 \times 10^{-3}$ and 10^{-2} ($\min y_j = -4.2 \times 10^{-5}$ and -2.9×10^{-5} , the size of $y_2^* = 3.65 \times 10^{-5}$) but not at $h \leq 2 \times 10^{-3}$. We did not see the effect as a visible oscillation in an error curve: its amplitude is comparable to the error of Implicit Euler, so E_∞ hides it (CN at $h = 10^{-2}$ has $E_\infty = 9.4 \times 10^{-5}$, below 1.04×10^{-4} for Implicit Euler); it shows only in the sign of y_2 .

6.3 Three-Fold Validation Strategy Recap

Our confidence in the reported numbers rests on three independent chains of evidence.

(i) **Independent oracle (Section 5.1).** The Radau reference agrees with a run at 100 times looser tolerances to 6.30×10^{-12} and with BDF to 1.94×10^{-11} at the 201 output times, and to 1.95×10^{-11} on a 60,000-point grid, which validates the dense output used for every E_∞ (Table 3).

(ii) **Known properties (Sections 5.4, 4.1 and 6.1).** Analytic facts are compared with computation: the invariant $y_1 + y_2 + y_3 = 1$ (Section 2.2) is preserved to round-off ($\sim 10^{-14}$) by the admissible runs we checked, which tests the implementation of f and of the linear algebra but not the accuracy, since every method used here preserves the invariant by construction; non-negativity and accuracy are therefore checked separately. The stability regions and the RK4 limit 2.7853 are computed to the digits shown (the limit as a numerical root of $|R(x)| = 1$); the initial-layer solution of Section 6.1 matches the reference to 3.7×10^{-5} ; and the stiffness ratio agrees with the Problem-Pack check values to their quoted digits (Table 1).

(iii) **Cross-order self-consistency (Section 5.3).** Methods of orders 1, 2 and 4 converge to the same reference with fitted slopes 1.06, 1.000 (first order), 2.004 (second) and 4.25 (fourth); the finest RK4 and CN runs, with errors 1.2×10^{-10} and 4.9×10^{-8} , agree with each other to within 5×10^{-8} . The RK4 slope slightly exceeds 4; presumably the reference uncertainty plays a role, since the finest RK4 error is only about six times 2×10^{-11} .

The prediction–observation loop of Section 4.3 (h_{crit} against h_{emp}) connects these chains to the stability claims. This is the chain of evidence on which we base the final numbers: $\mathbf{y}(40) \approx (0.715827069, 9.1855 \times 10^{-6}, 0.284163746)$ with no error claimed below 2×10^{-11} . The chain verifies the numerics, not the model: the oracle and all methods solve the same ODE, whose coefficients are fixed by the Problem Pack.

6.4 Limitations

- **Frozen-Jacobian analysis (Sections 4.1 and 4.3).** $h_{\text{crit}} = 2/|\lambda|_{\text{max}}$ is a local estimate along the *reference* trajectory, using real parts only; a numerical trajectory that has already left it (e.g. after the first negative value) sees other Jacobians. h_{emp} comes from a bisection that assumes a monotone transition, and the nine-time-unit delay between exceeding the local limit and divergence was measured on two runs only.
- **Non-normality (Section 4.2).** We computed only $\kappa(V)$ of the 2×2 reduced Jacobian, whose peak is a resolution artefact; the bound $\kappa(V) \max_j |1 + h\lambda_j|^n$ holds for a frozen matrix. We did not compute pseudospectra or the product of the step matrices along the trajectory, so transient growth is bounded only heuristically.

- **Newton iteration (Section 3.3).** The solver is a full Newton method with the analytic Jacobian, an Explicit-Euler predictor and an absolute residual test, without damping or positivity safeguards. Its convergence region was not analysed, only probed: a perturbed guess converges to a non-physical root, and a loose tolerance fails at $t \approx 3.71$ (Section 5.5.1). The rule $\tau_N \lesssim E_{\text{target}}/N_{\text{steps}}$ is a worst-case heuristic, and the value $\tau_N = 10^{-13}$ used for the order study meets it only with a small margin for the finest CN run and was not tested against looser ones.
- **Adaptive control (Section 3.5).** The explicit and implicit methods use an absolute max-norm proxy rather than component-scaled relative tolerances, so the small y_2 component is not controlled relatively. The extrapolated IE accepts a second-order state using a first-order error proxy; this is typically conservative in the asymptotic regime but is not an estimator of the extrapolated state’s own local error. Newton’s residual tolerance remains fixed at 10^{-13} rather than being coupled to each trial’s error budget. Non-negativity is monitored but not enforced, and the affine extrapolation does not preserve positivity by construction. The observed global ratio near one is specific to these runs, not a general tolerance guarantee.
- **Scope of the empirical conclusions.** All results concern one initial value problem, one parameter set and $T = 40$. Timings and the factor 3.2 come from one machine and one implementation, and the wall-clock crossover between methods is not reported for this reason; the reference certifies errors only down to $\sim 2 \times 10^{-11}$; and Tikhonov’s theorem guarantees only $O(\delta)$, whereas the QSSA is observed to be far more accurate, which we do not explain (Section 6.1).

7 Conclusions

We asked whether numerical solutions of the Robertson problem on $0 \leq t \leq 40$ can be trusted, and what they reveal about stiffness. Four self-implemented integrators (Explicit Euler, RK4, Implicit Euler and Trapezoidal/CN with Newton’s method) and adaptive Explicit and Implicit Euler methods (first-order and locally extrapolated IE variants) were compared with a tight Radau reference, whose discrepancy with BDF and with a looser run is about 2×10^{-11} . Our main findings are as follows.

1. **The results are trustworthy.** The implementations recover their formal orders (fitted slopes 1.06, 4.25, 1.000, 2.004), and three independent chains of evidence support the numbers (Section 6.3): $\mathbf{y}(40) \approx (0.715827069, 9.1855 \times 10^{-6}, 0.284163746)$.
2. **Stiffness is a stability restriction, not an accuracy restriction.** The fast mode dies within $t_f \approx 9.13 \times 10^{-4}$ and y_2 is then slaved to a quasi-steady value (Section 6.1), yet it keeps $|\lambda|_{\text{max}}$ at $O(10^3)$, not 3×10^7 . The frozen-Jacobian estimate $h_{\text{crit}} = 5.895 \times 10^{-4}$ predicts the Explicit-Euler boundary $h_{\text{emp}} = 5.962 \times 10^{-4}$ to 1.1%; RK4 is limited to 8.21×10^{-4} , only about 1.4 times more, and explicit step-size adaptivity does not remove the limit. Our adaptive IE takes much larger steps after the initial layer.
3. **Implicit methods succeed, with a caveat.** L-stable Implicit Euler reaches $E_\infty \approx 10^{-4}$ with $h = 10^{-2}$, about seventeen times h_{crit} . Trapezoidal/CN is A-stable but not L-stable, and at the tested steps $h = 5 \times 10^{-3}$ and 10^{-2} it loses non-negativity, in line with the loss of damping ($R \rightarrow -1$) that the stability analysis predicts.
4. **Diagnostics must be kept apart.** Conservation stays at round-off even when the solution is negative or inaccurate (all methods preserve $\mathbf{1}^T \mathbf{y}$ by construction, since $\mathbf{1}^T \mathbf{f} \equiv 0$ and, for the Newton updates, $\mathbf{1}^T \mathbf{J} \equiv 0$); a small Newton residual can belong to a non-physical root; tolerances must resolve $y_2 \sim 10^{-5}$.

5. **Efficiency depends on the accuracy demanded.** Our extrapolated adaptive IE reaches $E_\infty = 9.00 \times 10^{-7}$ in 501 accepted steps at $\text{tol} = 1 \times 10^{-6}$, compared with 25 398 steps and 1.97×10^{-5} error for adaptive EE. The first-order IE comparison shows that extrapolation supplies an additional accuracy gain. The work–precision curves retain RK4 and CN as competitive higher-order choices at tight achieved errors, subject to non-negativity and machine-dependent timing. SciPy Radau remains a separate library baseline, with 66 steps and $E_\infty \approx 6.0 \times 10^{-8}$.

Practical recommendation. For Robertson-type problems, use an L-stable implicit or a stiff adaptive solver, choose the Newton tolerance well below the smallest relevant component, and monitor non-negativity and error in addition to conservation. On the short horizon $T = 40$ studied here, a higher-order explicit method such as RK4 remains competitive when the accuracy target is tight enough that the step is limited by accuracy rather than stability.

Limitations and outlook. The conclusions are empirical and restricted to one problem, one machine and a frozen-Jacobian analysis (Section 6.4). Natural extensions are component-scaled absolute/relative error control, Newton tolerances coupled to the local error budget, embedded higher-order stiff schemes, positivity-preserving integration, and a pseudospectral study of non-normality. Testing several stiff problems and longer integration horizons would assess how far the present adaptive results generalise.

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9 Appendix

Appendix A: AI Transparency Log

Date/Step	AI Prompt (summary)	What we changed & why	Verification
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Table 11: AI Transparency Log.

Appendix B: Individual Contribution Statement (ICS)

This statement was written by the project manager and agreed by the whole team.

Member	Role	Main responsibilities
Rengerile Na	Project Manager	coordination, mathematical formulation, mutual cross-checking of results, ...
Suning Peng	Algorithm Implementation	solvers, step-size experiment, sanity check, mutual cross-checking of results, ...
Kainan Zhao	Testing & Validation	reference solution, diagnostics, efficiency comparsion, references, mutual cross-checking of results, ...
Yixun Wu	Visualization & Report	the framework of the report and designed the slide, introduction, conclusion, ...
Keji Chen	Mathematical Theory	numerical integration methods, stability regions of four methods, QSSA, ...

Rengerile Na. Responsible for the section 2.1, 2.2, 2.3, 4.2, 6.2, 6.3, 6.4, after-class coordination(hold meeting, run git repository, ...), mutual cross-checking of results.

Suning Peng. Responsible for the section 3.5, 4.3, 5.2, 5.3, 5.4, mutual cross-checking of results, integrating all the code and writing the README.

Kainan Zhao. Responsible for the section 5.1 5.5, 5.6, mutual cross-checking of results, finding all the support references.

Yixun Wu. Responsible for the section 1.1, 1.2, 7, drafting the framework of the whole report and designing the template of the slide in Latex.

Keji Chen. Responsible for the section 3.1, 3.2, 3.3, 3.4, 4.1, 6.1.

Shared work. Weekly meetings, joint editing of the final report, preparation of the seminar, presentation.